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Gas turbine engine with forward swept outlet guide vanes

Abstract

A turbofan engine defining an axial direction and a longitudinal centerline along the axial direction is provided. The turbofan engine includes: a fan section having a fan, the fan comprising a plurality of fan blades; a turbomachine drivingly coupled to the fan, the turbomachine comprising a compressor section with a low pressure compressor, a turbine section with a low pressure turbine, a reduction gearbox, and an outer casing, the low pressure turbine drivingly coupled to the low pressure compressor across the reduction gearbox; an outer nacelle surrounding the fan and at least a portion of the turbomachine; an outlet guide vane extending between the turbomachine and the outer nacelle at a location downstream of the plurality of fan blades, the outlet guide vane defining a base and a tip and being forward swept from the base to the tip.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2634581	12/1952	Kent	N/A	N/A
2663993	12/1952	Mosser	N/A	N/A
3533486	12/1969	Paulson	N/A	N/A
3540682	12/1969	Dibble et al.	N/A	N/A
3575259	12/1970	Wilkinson	N/A	N/A
3618876	12/1970	Skidmore et al.	N/A	N/A
3664612	12/1971	Skidmore et al.	N/A	N/A
3819008	12/1973	Evans et al.	N/A	N/A
3861822	12/1974	Wanger	N/A	N/A
3981466	12/1975	Shah	N/A	N/A
4010608	12/1976	Simmons	N/A	N/A
4254619	12/1980	Griffin, III et al.	N/A	N/A
4548034	12/1984	Maguire	N/A	N/A
4688745	12/1986	Rosenthal Herman A.	N/A	N/A
4916894	12/1989	Adamson et al.	N/A	N/A
4940386	12/1989	Feuvrier	415/189	F01D 9/042
5010729	12/1990	Adamson et al.	N/A	N/A
5012420	12/1990	Walker et al.	N/A	N/A
5458457	12/1994	Goto et al.	N/A	N/A
5642985	12/1996	Spear et al.	N/A	N/A
5791136	12/1997	Utamura et al.	N/A	N/A
6195983	12/2000	Wadia et al.	N/A	N/A
6409469	12/2001	Tse	N/A	N/A
6431820	12/2001	Beacock et al.	N/A	N/A
6457938	12/2001	Liu et al.	N/A	N/A
6508630	12/2002	Liu et al.	N/A	N/A
6540477	12/2002	Glynn et al.	N/A	N/A
6546734	12/2002	Antoine et al.	N/A	N/A
6585482	12/2002	Liotta et al.	N/A	N/A
6655632	12/2002	Gupta et al.	N/A	N/A
6834505	12/2003	Al-Roub et al.	N/A	N/A
6935833	12/2004	Seitz	N/A	N/A

7025565	12/2005	Urso et al.	N/A	N/A
7055304	12/2005	Courtot et al.	N/A	N/A
7210900	12/2006	Urso et al.	N/A	N/A
7246773	12/2006	Stoner et al.	N/A	N/A
7313963	12/2007	Kuznar	N/A	N/A
7374403	12/2007	Decker et al.	N/A	N/A
7374404	12/2007	Schilling	N/A	N/A
7407364	12/2007	Arnold et al.	N/A	N/A
7413401	12/2007	Szucs et al.	N/A	N/A
7493754	12/2008	Moniz et al.	N/A	N/A
7514810	12/2008	Kern et al.	N/A	N/A
7575412	12/2008	Seitz	N/A	N/A
7594388	12/2008	Cherry et al.	N/A	N/A
7694505	12/2009	Schilling	N/A	N/A
7716914	12/2009	Schilling	N/A	N/A
7789620	12/2009	Vontell, Sr. et al.	N/A	N/A
7849669	12/2009	Keogh	N/A	N/A
7854778	12/2009	Groom et al.	N/A	N/A
7871244	12/2010	Marini et al.	N/A	N/A
7874137	12/2010	Chaney et al.	N/A	N/A
7882694	12/2010	Suciu et al.	N/A	N/A
7926261	12/2010	Porte	N/A	N/A
7955046	12/2010	McCune et al.	N/A	N/A
7966806	12/2010	Henry et al.	N/A	N/A
8021104	12/2010	Gu et al.	N/A	N/A
8105042	12/2011	Parkin et al.	N/A	N/A
8186942	12/2011	Haas	N/A	N/A
8226360	12/2011	Scoggins et al.	N/A	N/A
8328518	12/2011	Liang et al.	N/A	N/A
8375695	12/2012	Schilling et al.	N/A	N/A
8397487	12/2012	Sennoun et al.	N/A	N/A
8419374	12/2012	Huth et al.	N/A	N/A
8529188	12/2012	Winter	N/A	N/A
8572943	12/2012	Sheridan	N/A	N/A
8747055	12/2013	McCune et al.	N/A	N/A
8677764	12/2013	Porte	N/A	N/A
8689538	12/2013	Sankrithi et al.	N/A	N/A
8690519	12/2013	Aalburg et al.	N/A	N/A
8757959	12/2013	Suciu et al.	N/A	N/A
8806871	12/2013	McMasters et al.	N/A	N/A
8899915	12/2013	McCune et al.	N/A	N/A
8943796	12/2014	McCaffrey	N/A	N/A
8956108	12/2014	Eleftheriou et al.	N/A	N/A
9003808	12/2014	Kupratis	N/A	N/A
9039364	12/2014	Alvanos et al.	N/A	N/A
9045991	12/2014	Read et al.	N/A	N/A
9068470	12/2014	Mills et al.	N/A	N/A
9074483	12/2014	Breeze-Stringfellow et al.	N/A	N/A
9109514	12/2014	Cheong	N/A	N/A

9114877	12/2014	Weber et al.	N/A	N/A
9140212	12/2014	Moon et al.	N/A	N/A
9206697	12/2014	Tibbott et al.	N/A	N/A
9206740	12/2014	Wong et al.	N/A	N/A
9222417	12/2014	Kupratis et al.	N/A	N/A
9228497	12/2015	Ottow et al.	N/A	N/A
9239005	12/2015	Strecker et al.	N/A	N/A
9506361	12/2015	Fielding et al.	N/A	N/A
9523284	12/2015	Miller et al.	N/A	N/A
9540094	12/2016	Negulescu et al.	N/A	N/A
9695751	12/2016	Kupratis et al.	N/A	N/A
9784134	12/2016	Eleftheriou	N/A	N/A
9797261	12/2016	Tibbott et al.	N/A	N/A
9835092	12/2016	Sawyers-Abbott et al.	N/A	N/A
9840930	12/2016	Lee et al.	N/A	N/A
9863256	12/2017	Lee et al.	N/A	N/A
9874102	12/2017	Azad et al.	N/A	N/A
9874109	12/2017	Hatcher, Jr. et al.	N/A	N/A
9879599	12/2017	Mercier et al.	N/A	N/A
9945247	12/2017	Appukuttan et al.	N/A	N/A
9957918	12/2017	Suciu et al.	N/A	N/A
9995167	12/2017	Shepard	N/A	N/A
10018167	12/2017	Tentorio	N/A	N/A
10054052	12/2017	Zheng et al.	N/A	N/A
10060270	12/2017	Lee et al.	N/A	N/A
10060351	12/2017	Oggero	N/A	N/A
10066508	12/2017	Geiger	N/A	N/A
10100733	12/2017	O'Toole et al.	N/A	N/A
10113444	12/2017	Huang et al.	N/A	N/A
10131443	12/2017	Namgoong	N/A	N/A
10138817	12/2017	Venter	N/A	N/A
10145301	12/2017	Abrari et al.	N/A	N/A
10173780	12/2018	Mackin et al.	N/A	N/A
10184340	12/2018	Baltas et al.	N/A	N/A
10189572	12/2018	Mackin et al.	N/A	N/A
10196895	12/2018	Weinert et al.	N/A	N/A
10196901	12/2018	Wong et al.	N/A	N/A
10215192	12/2018	Griffin	N/A	N/A
10239626	12/2018	Herchenroder et al.	N/A	N/A
10240526	12/2018	Suciu et al.	N/A	N/A
10252790	12/2018	Ramakrishnan	N/A	N/A
10260527	12/2018	Steen	N/A	N/A
10273880	12/2018	Kolvick et al.	N/A	N/A
10288010	12/2018	Houston et al.	N/A	N/A
10288083	12/2018	Miller et al.	N/A	N/A
10344711	12/2018	Hsu	N/A	N/A
10371054	12/2018	Sasse et al.	N/A	N/A
10378554	12/2018	Yu et al.	N/A	N/A
10385871	12/2018	Lurie et al.	N/A	N/A

10393132	12/2018	Lee et al.	N/A	N/A
10399664	12/2018	Bowden et al.	N/A	N/A
10415409	12/2018	Oyarbide	N/A	N/A
10436046	12/2018	Fentem et al.	N/A	N/A
10480328	12/2018	Weaver et al.	N/A	N/A
10480413	12/2018	Snyder	N/A	N/A
10502232	12/2018	Mohtar et al.	N/A	N/A
10519859	12/2018	Marchaj et al.	N/A	N/A
10557415	12/2019	Boudebiza et al.	N/A	N/A
10563513	12/2019	Kalitzin et al.	N/A	N/A
10634059	12/2019	Rami et al.	N/A	N/A
10641178	12/2019	Corrandini et al.	N/A	N/A
10690146	12/2019	Urac et al.	N/A	N/A
10695704	12/2019	Mook et al.	N/A	N/A
10711797	12/2019	Kroger et al.	N/A	N/A
10724395	12/2019	Kupratis et al.	N/A	N/A
10724435	12/2019	Kroger et al.	N/A	N/A
10738694	12/2019	Kupratis et al.	N/A	N/A
10787909	12/2019	Hiernaux	N/A	N/A
10794224	12/2019	Schiessl	N/A	N/A
10794292	12/2019	Kupratis et al.	N/A	N/A
10794395	12/2019	Tamada	N/A	N/A
10794396	12/2019	Kroger et al.	N/A	N/A
10801442	12/2019	Clements et al.	N/A	N/A
10815886	12/2019	Kroger et al.	N/A	N/A
10816208	12/2019	DiCintio et al.	N/A	N/A
10822999	12/2019	Morris et al.	N/A	N/A
10823114	12/2019	Clements et al.	N/A	N/A
10830135	12/2019	Slawinska et al.	N/A	N/A
10837361	12/2019	Abrari et al.	N/A	N/A
10883515	12/2020	Lurie et al.	N/A	N/A
10961864	12/2020	Miranda et al.	N/A	N/A
11085515	12/2020	Moniz et al.	N/A	N/A
11118601	12/2020	Yu et al.	N/A	N/A
11401824	12/2021	Breeze-Stringfellow et al.	N/A	N/A
11421627	12/2021	Moniz et al.	N/A	N/A
11480063	12/2021	Miller et al.	N/A	N/A
11661851	12/2022	Yao	415/191	F02K 3/06
11725526	12/2022	Sibbach et al.	N/A	N/A
11746664	12/2022	Muldoon et al.	N/A	N/A
2001/0023582	12/2000	Nagel	N/A	N/A
2002/0197156	12/2001	Haller	N/A	N/A
2005/0201856	12/2004	Koshoffer	N/A	N/A
2006/0272314	12/2005	Moniz et al.	N/A	N/A
2008/0041064	12/2007	Moore et al.	N/A	N/A
2008/0155959	12/2007	Rasheed et al.	N/A	N/A
2008/0155989	12/2007	Roth	N/A	N/A
2008/0232963	12/2007	Durocher et al.	N/A	N/A
2009/0155053	12/2008	Speak	N/A	N/A

2009/0155072	12/2008	Winter	N/A	N/A
2009/0245997	12/2008	Hurwitz et al.	N/A	N/A
2009/0255116	12/2008	McMasters et al.	N/A	N/A
2009/0272850	12/2008	Rainous et al.	N/A	N/A
2009/0277181	12/2008	Druon et al.	N/A	N/A
2009/0293445	12/2008	Ress, Jr.	N/A	N/A
2011/0044796	12/2010	Hussain et al.	N/A	N/A
2012/0198816	12/2011	Suciu	60/226.3	B64D 27/40
2012/0240594	12/2011	Shamara	N/A	N/A
2012/0241561	12/2011	Shamara	N/A	N/A
2013/0186058	12/2012	Sheridan	60/226.1	F01D 25/168
2013/0192263	12/2012	Suciu et al.	N/A	N/A
2013/0223974	12/2012	Schwarz et al.	N/A	N/A
2013/0319011	12/2012	Pescosolido et al.	N/A	N/A
2013/0323011	12/2012	Chopra et al.	N/A	N/A
2014/0020404	12/2013	Sheridan et al.	N/A	N/A
2014/0037443	12/2013	Khan	N/A	N/A
2014/0208755	12/2013	Ekanayake et al.	N/A	N/A
2014/0208768	12/2013	Bacic	N/A	N/A
2014/0318137	12/2013	McMasters et al.	N/A	N/A
2015/0176486	12/2014	Menheere et al.	N/A	N/A
2015/0198050	12/2014	Lee et al.	N/A	N/A
2015/0198163	12/2014	Lei et al.	N/A	N/A
2015/0308353	12/2014	Gardner et al.	N/A	N/A
2016/0003046	12/2015	Smith et al.	N/A	N/A
2016/0017751	12/2015	Caruel	N/A	N/A
2016/0047307	12/2015	Williamson et al.	N/A	N/A
2016/0047308	12/2015	Williamson et al.	N/A	N/A
2016/0047309	12/2015	Davidson et al.	N/A	N/A
2016/0047335	12/2015	Davidson et al.	N/A	N/A
2016/0084265	12/2015	Yu et al.	N/A	N/A
2016/0195010	12/2015	Roberge	N/A	N/A
2016/0298543	12/2015	Suciu et al.	N/A	N/A
2017/0030213	12/2016	Vlasic et al.	N/A	N/A
2017/0082028	12/2016	Duong et al.	N/A	N/A
2017/0138202	12/2016	Wadia et al.	N/A	N/A
2017/0146026	12/2016	Griffin	N/A	N/A
2017/0190438	12/2016	Qiu	N/A	B64D 29/00
2017/0191372	12/2016	Tralshawala et al.	N/A	N/A
2017/0276018	12/2016	Bifulco et al.	N/A	N/A
2017/0284220	12/2016	Roberge	N/A	N/A
2017/0292532	12/2016	Wall	N/A	N/A
2017/0297728	12/2016	Niergarth et al.	N/A	N/A
2017/0298751	12/2016	Messmann et al.	N/A	N/A
2017/0314509	12/2016	Laricchiuta et al.	N/A	N/A
2017/0342839	12/2016	Miller et al.	N/A	N/A
2018/0010617	12/2017	Casavant et al.	N/A	N/A

2018/0045059	12/2017	Lee et al.	N/A	N/A
2018/0094582	12/2017	Rosenau et al.	N/A	N/A
2018/0112546	12/2017	Griffin	N/A	N/A
2018/0128179	12/2017	Staubach et al.	N/A	N/A
2018/0135432	12/2017	Martin, Jr. et al.	N/A	N/A
2018/0138766	12/2017	Moore et al.	N/A	N/A
2018/0138767	12/2017	Moore et al.	N/A	N/A
2018/0216527	12/2017	D'Angelo et al.	N/A	N/A
2018/0223733	12/2017	Zhang et al.	N/A	N/A
2018/0230912	12/2017	Hasel	N/A	F02C 7/36
2018/0230944	12/2017	Suciu et al.	N/A	N/A
2018/0283183	12/2017	Gallier et al.	N/A	N/A
2018/0334916	12/2017	Lurie et al.	N/A	N/A
2018/0363554	12/2017	Kroger et al.	N/A	N/A
2018/0363675	12/2017	Kroger et al.	N/A	N/A
2018/0363676	12/2017	Kroger et al.	N/A	N/A
2018/0363677	12/2017	Kroger et al.	N/A	N/A
2018/0363678	12/2017	Kroger et al.	N/A	N/A
2019/0010868	12/2018	Davidson et al.	N/A	N/A
2019/0048798	12/2018	Slawinska et al.	N/A	N/A
2019/0063243	12/2018	Lemaire	N/A	N/A
2019/0107119	12/2018	Yu et al.	N/A	N/A
2019/0226402	12/2018	Kray et al.	N/A	N/A
2019/0284693	12/2018	Task	N/A	N/A
2019/0284940	12/2018	Task et al.	N/A	N/A
2019/0338707	12/2018	Beecroft et al.	N/A	N/A
2019/0368421	12/2018	Sweidan	N/A	N/A
2020/0025110	12/2019	Silkowski	N/A	N/A
2020/0049077	12/2019	Schwarz et al.	N/A	N/A
2020/0080432	12/2019	Filipenco	N/A	N/A
2020/0123918	12/2019	Duong et al.	N/A	N/A
2020/0141277	12/2019	Cooper et al.	N/A	N/A
2020/0149421	12/2019	Lad	N/A	N/A
2020/0165938	12/2019	Schreiber et al.	N/A	N/A
2020/0217272	12/2019	Gangloff, Jr.	N/A	F23M 20/005
2020/0276641	12/2019	Benard et al.	N/A	N/A
2020/0284261	12/2019	Merchant et al.	N/A	N/A
2020/0300174	12/2019	Williamson et al.	N/A	N/A
2020/0316508	12/2019	Mook et al.	N/A	N/A
2020/0347737	12/2019	Bordoni	N/A	N/A
2021/0036392	12/2020	Taykov et al.	N/A	N/A
2021/0078715	12/2020	Bevirt et al.	N/A	N/A
2021/0087948	12/2020	Suciu	N/A	N/A
2021/0108597	12/2020	Ostdiek et al.	N/A	N/A
2021/0363921	12/2020	Thomas et al.	N/A	N/A
2023/0028503	12/2022	Sibbach et al.	N/A	N/A
2023/0053554	12/2022	Sibbach et al.	N/A	N/A
2023/0060010	12/2022	Sibbach et al.	N/A	N/A
2023/0119477	12/2022	Miller et al.	N/A	N/A

2023/0250723	12/2022	Kuropatwa et al.	N/A	N/A
2023/0265862	12/2022	Sibbach et al.	N/A	N/A
2023/0279872	12/2022	Miller et al.	N/A	N/A
2023/0287837	12/2022	Sibbach et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101922312	12/2009	CN	N/A
2508711	12/2011	EP	N/A
2617965	12/2012	EP	N/A
1038694	12/1952	FR	N/A
722001	12/1954	GB	N/A
2403778	12/2004	GB	N/A
2594712	12/2020	GB	F01D 9/041

OTHER PUBLICATIONS

Willis, Quiet Clean Short-Haul Experimental Engine (QCSEE) Final Report, NASA-CR-159473, National Aeronautics and Space Administration, Contract NAS3-18021, QCSEE NASA Report NASA-CR-159473, Aug. 1979, 312 Pages. cited by applicant

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Background/Summary

FIELD
(1) The present disclosure relates to a gas turbine engine having forward swept outlet guide vanes.
BACKGROUND
(2) A gas turbine engine generally includes a turbomachine and a rotor assembly. Gas turbine engines, such as turbofan engines, may be used for aircraft propulsion. In the case of a turbofan engine, the rotor assembly may be configured as a fan assembly, and the turbofan engine may include an outer nacelle surrounding at least in part a fan of the fan assembly and a turbomachine configured to drive the fan.
(3) The outer nacelle may be coupled to the turbomachine at least in part by a plurality of outlet guide vanes. The plurality of outlet guide vanes may be located downstream of the fan and operate to straighten an airflow from the fan during operation of the turbofan engine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:
(2) FIG. **1** is a cross-sectional view of a gas turbine engine in accordance with an exemplary aspect of the present disclosure.
(3) FIG. **2** is a close-up view of a portion of the exemplary gas turbine engine of FIG. **1**.

(4) FIG. **3** is a schematic, axial view of a portion of an outer casing of a turbomachine of the exemplary gas turbine engine of FIG. **1**.
(5) FIG. **4** is another schematic, axial view of a portion of the outer casing of the turbomachine of the exemplary gas turbine engine of FIG. **1**.
(6) FIG. **5** is a schematic view of an acoustic treatment in accordance with an exemplary aspect of the present disclosure.
(7) FIG. **6** is a schematic view of an acoustic treatment in accordance with another exemplary aspect of the present disclosure.
(8) FIG. **7** is a schematic view of an acoustic treatment in accordance with yet another exemplary aspect of the present disclosure.
(9) FIG. **8** is a schematic view of an acoustic treatment in accordance with still another exemplary aspect of the present disclosure.
(10) FIG. **9** is a cross-sectional view of a gas turbine engine in accordance with another exemplary aspect of the present disclosure.
(11) FIG. **10** is a perspective view of a portion of the exemplary gas turbine engine of FIG. **9**.
(12) FIG. **11** is an axial view of a portion of the exemplary gas turbine engine of FIG. **9**.
(13) FIG. **12** is a cross-sectional view of a gas turbine engine in accordance with yet another exemplary aspect of the present disclosure.
(14) FIG. **13** is a cross-sectional view of a gas turbine engine in accordance with still another exemplary aspect of the present disclosure.
(15) FIG. **14** is a cross-sectional view of a gas turbine engine in accordance with yet another exemplary aspect of the present disclosure.
(16) FIG. **15** is a close-up view of a portion of the exemplary gas turbine engine of FIG. **14**.
(17) FIG. **16** is an axial view of a gas turbine engine in accordance with yet an exemplary aspect of the present disclosure.
(18) FIG. **17** is an axial view of a gas turbine engine in accordance with yet another exemplary aspect of the present disclosure.
(19) FIG. **18** is a first cross-sectional view of an inlet pre-swirl feature depicted in FIGS. **14** and **15**.
(20) FIG. **19** is a second cross-sectional view of the inlet pre-swirl feature depicted in FIGS. **14** and **15**.
(21) FIG. **20** is a close-up view of a portion of a gas turbine engine in accordance with yet another exemplary aspect of the present disclosure.
(22) FIG. **21** is an axial view of a gas turbine engine in accordance with yet another exemplary aspect of the present disclosure.
(23) FIG. **22** is a schematic view of a controller in accordance with an exemplary embodiment of the present disclosure.
(24) FIG. **23** is a cross-sectional view of a gas turbine engine in accordance with yet another exemplary aspect of the present disclosure.

DETAILED DESCRIPTION

(25) Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.
(26) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.
(27) The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

(28) The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C.
(29) The phrases “from X to Y” and “between X and Y” each refers to a range of values inclusive of the endpoints (i.e., refers to a range of values that includes both X and Y).
(30) The term “turbomachine” refers to a machine including one or more compressors, a heat generating section (e.g., a combustion section), and one or more turbines that together generate a torque output.
(31) The term “gas turbine engine” refers to an engine having a turbomachine as all or a portion of its power source. Example gas turbine engines include turbofan engines, turboprop engines, turbojet engines, turboshaft engines, etc., as well as hybrid-electric versions of one or more of these engines.
(32) The term “combustion section” refers to any heat addition system for a turbomachine. For example, the term combustion section may refer to a section including one or more of a deflagrative combustion assembly, a rotating detonation combustion assembly, a pulse detonation combustion assembly, or other appropriate heat addition assembly. In certain example embodiments, the combustion section may include an annular combustor, a can combustor, a cannular combustor, a trapped vortex combustor (TVC), or other appropriate combustion system, or combinations thereof.
(33) The terms “low” and “high”, or their respective comparative degrees (e.g., -er, where applicable), when used with a compressor, a turbine, a shaft, or spool components, etc. each refer to relative speeds within an engine unless otherwise specified. For example, a “low turbine” or “low speed turbine” defines a component configured to operate at a rotational speed, such as a maximum allowable rotational speed, lower than a “high turbine” or “high speed turbine” of the engine.
(34) The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle, and refer to the normal operational attitude of the gas turbine engine or vehicle. For example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.
(35) The term “bypass ratio” refers to a ratio in an engine of an amount of airflow that is bypassed around the engine's ducted inlet to the amount that passes through the engine's ducted inlet. For example, in the embodiment of FIG. **1**, discussed below, the bypass ratio refers to an amount of airflow from the fan that flows over the outer casing to an amount of airflow from the fan that flows through the engine inlet.
(36) The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.
(37) As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.
(38) For purposes of the description hereinafter, the terms “upper”, “lower”, “right”, “left”, “vertical”, “horizontal”, “top”, “bottom”, “lateral”, “longitudinal”, and derivatives thereof shall relate to the embodiments as they are oriented in the drawing figures. However, it is to be understood that the embodiments may assume various alternative variations, except where expressly specified to the contrary. It is also to be understood that the specific devices illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the disclosure. Hence, specific dimensions and other physical characteristics related to the embodiments disclosed herein are not to be considered as limiting.
(39) Turbofan engine design continues to push for more efficient and compact engines. Traditionally, the design of turbofan engines has leaned towards the inclusion of radially-oriented or aft-swept Outlet Guide Vanes (OGVs). This design decision has been influenced by the need to minimize acoustic emissions and to keep the OGVs at a distance spaced from the fan blade tips.

While these designs may address acoustic concerns, they require an elongated turbofan engine and outer nacelle. A longer outer nacelle can impact the aerodynamic performance and efficiency of the turbofan engine.
(40) Contrary to the conventional turbofan engine design, the inventors of the present disclosure have found that inclusion of forward-swept OGVs can allow for desired shortening of the turbofan engine and outer nacelle, and inclusion of specific acoustic treatments on an outer casing of a turbomachine of the turbofan engine can address the specific acoustic concerns of such a design. In particular, inclusion of specific acoustic treatments on the outer casing may allow for the engine to remain acoustically compliant while benefiting from the structural and aerodynamic advantages of the forward-swept design. Such a configuration may therefore provide for improved aerodynamics and efficiency, attributes highly desired in the aerospace sector.
(41) Referring now to the drawings, wherein identical numerals indicate the same elements throughout the figures, FIG. **1** is a schematic cross-sectional view of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure. More particularly, for the embodiment of FIG. **1**, the gas turbine engine is a high-bypass turbofan jet engine, sometimes also referred to as a "turbofan engine **10**." As shown in FIG. **1**, the turbofan engine **10** defines an axial direction A (extending parallel to a longitudinal centerline **12** provided for reference), a radial direction R, and a circumferential direction C extending about the longitudinal centerline **12**. In general, the turbofan engine **10** includes a fan section **14** and a turbomachine **16** disposed downstream from the fan section **14**.
(42) The exemplary turbomachine **16** depicted generally includes a substantially tubular outer casing **18** that defines an annular inlet **20**. The outer casing **18** encases, in serial flow relationship, a compressor section including a booster or low pressure (LP) compressor **22** and a high pressure (HP) compressor **24**; a combustion section **26**; a turbine section including a high pressure (HP) turbine **28** and a low pressure (LP) turbine **30**; and a jet exhaust nozzle section **32**. A high pressure (HP) shaft **34** (which may additionally or alternatively be a spool) drivingly connects the HP turbine **28** to the HP compressor **24**. A low pressure (LP) shaft **36** (which may additionally or alternatively be a spool) drivingly connects the LP turbine **30** to the LP compressor **22**. The compressor section, combustion section **26**, turbine section, and jet exhaust nozzle section **32** together define a working gas flowpath **37**.
(43) For the embodiment depicted, the fan section **14** includes a fan **38** having a plurality of fan blades **40** coupled to a disk **42** in a spaced apart manner. The fan **38** more specifically includes a single stage of the fan blades **40**, and thus may be referred to as a single stage fan. As depicted, the fan blades **40** extend outwardly from disk **42** generally along the radial direction R and define a fan diameter D, along the radial direction R (the fan diameter D equal to twice a fan radius **45**, depicted in FIG. **1**). In at least certain exemplary aspects, the fan diameter D may be greater than or equal to 4 feet and less than or equal to 18 feet. For example, in certain exemplary aspects, the fan diameter D may be greater than or equal to 5 feet, such as greater than or equal to 6 feet, such as greater than or equal to 8 feet.
(44) The turbofan engine **10** further includes a power gearbox **46**, and the fan blades **40** and disk **42** are together rotatable about the longitudinal centerline **12** by LP shaft **36** across the power gearbox **46**. The power gearbox **46** includes a plurality of gears for adjusting a rotational speed of the fan **38** relative to a rotational speed of the LP shaft **36**, such that the fan **38** may rotate at a more efficient fan speed. In certain exemplary embodiments, the power gearbox **46** may define a gear ratio of at least 2:1, such as at least 4:1, and less than or equal to 12:1.
(45) Referring still to the exemplary embodiment of FIG. **1**, the disk **42** is covered by a rotatable front hub **48** of the fan section **14** (sometimes also referred to as a "spinner"). The front hub **48** is aerodynamically contoured to promote an airflow through the plurality of fan blades **40**.
(46) Additionally, the exemplary fan section **14** includes an annular fan casing or outer nacelle **50** that circumferentially surrounds the fan **38** and/or at least a portion of the turbomachine **16**. It

should be appreciated that the outer nacelle **50** is supported relative to the turbomachine **16** by a plurality of circumferentially-spaced outlet guide vanes **52**. The outer nacelle **50** defines a bypass airflow passage **56** with the turbomachine **16**. As will be discussed in more detail below, the plurality of outlet guide vanes **52** are forward swept.
(47) As is depicted schematically, the outer nacelle **50** houses a thrust reverser assembly **54** that may be employed during, e.g., landing operations, to help slow down an aircraft. Aft of the thrust reverser assembly **54**, the outer nacelle **50** includes a closeout section **55** aerodynamically shaped to reduce a drag on the outer nacelle **50**.
(48) During operation of the turbofan engine **10**, a volume of air **58** enters the turbofan engine **10** through an associated inlet **60** of the nacelle **50** and fan section **14**. As the volume of air **58** passes across the fan blades **40**, a first portion of air as indicated by arrow **62** is directed or routed into the bypass airflow passage **56** and a second portion of air as indicated by arrow **64** is directed or routed into the working gas flowpath **37**, or more specifically into the LP compressor **22**. The ratio between the first portion of air **62** and the second portion of air **64** is commonly known as a bypass ratio. Notably, in the present exemplary embodiment, the bypass ratio of the turbofan engine **10** may be at least 4:1, such as at least 5:1, such as at least 8:1, such as less than or equal to 20:1.
(49) A pressure of the second portion of air **64** is increased as it is routed through the HP compressor **24** and into the combustion section **26**, where it is mixed with fuel and burned to provide combustion gases.
(50) The combustion gases are routed through the HP turbine **28** where a portion of thermal and/or kinetic energy from the combustion gases is extracted via sequential stages of HP turbine stator vanes that are coupled to the outer casing **18** and HP turbine rotor blades that are coupled to the HP shaft **34**, thus causing the HP shaft **34** to rotate, thereby supporting operation of the HP compressor **24**. The combustion gases are then routed through the LP turbine **30** where a second portion of thermal and kinetic energy is extracted from the combustion gases via sequential stages of LP turbine stator vanes that are coupled to the outer casing **18** and LP turbine rotor blades that are coupled to the LP shaft **36**, thus causing the LP shaft **36** to rotate, thereby supporting operation of the LP compressor **22** and/or rotation of the fan **38**.
(51) The combustion gases are subsequently routed through the jet exhaust nozzle section **32** of the turbomachine **16** to provide propulsive thrust. Simultaneously, the pressure of the first portion of air (indicated at arrow **62**) is substantially increased as the first portion of air is routed through the bypass airflow passage **56** before it is exhausted from a fan nozzle exhaust section **66** of the turbofan engine **10**, also providing propulsive thrust.
(52) In some exemplary embodiments, the exemplary turbofan engine **10** of the present disclosure may be a relatively large power class turbofan engine **10**. Accordingly, when operated in a high power operating condition (e.g., takeoff), the turbofan engine **10** may be configured to generate a relatively large amount of thrust. More specifically, when operated in the high power operating condition, the turbofan engine **10** may be configured to generate at least about 20,000 pounds of thrust, such as at least about 25,000 pounds of thrust, such as at least about 30,000 pounds of thrust, and less than or equal to, e.g., about 150,000 pounds of thrust. Accordingly, the turbofan engine **10** may be referred to as a relatively large power class turbofan engine **10**.
(53) It should be appreciated, however, that the exemplary turbofan engine **10** depicted in FIG. **1** is by way of example only, and that in other exemplary embodiments, the turbofan engine **10** may have any other suitable configuration. For example, although the turbofan engine **10** depicted includes a fan **38** configured as a fixed pitch fan, in other embodiments, the turbofan engine **10** may alternatively include a fan having fan blades that are rotatable about a pitch axis by, e.g., a pitch change mechanism.
(54) As will be appreciated, the turbofan engine **10** of the present disclosure is designed to reduce an overall length of the turbofan engine **10**, improving various aerodynamic aspects of the turbofan engine **10**, improving an overall efficiency of the turbofan engine **10**, and improving various

packaging concerns for the turbofan engine **10**.
(55) In particular, the turbofan engine **10** is provided with a reduced overall length at least in part due to the plurality of outlet guide vanes **52**. In particular, the plurality of outlet guide vanes **52** includes an outlet guide vane **52** extending between the turbomachine **16** and the outer nacelle **50**, defining a base **70** at an inner end along the radial direction R and a tip **72** at an outer end along the radial direction R, respectively. Contrary to conventional design understandings, the outlet guide vane **52** is forward swept from the base **70** to the tip **72**.
(56) More specifically, referring now also to FIG. **2**, providing a close-up view of the outlet guide vane **52** of FIG. **1**, the outlet guide vane **52** defines an OGV reference line **74** extending from an inner junction **76** between the outlet guide vane **52** and the turbomachine **16** at a leading edge **80** of the outlet guide vane **52**, and an outer junction **78** between the outlet guide vane **52** and the outer nacelle **50** at the leading edge **80** of the outlet guide vane **52**. The turbofan engine **10** further defines a radial reference line **82** extending perpendicularly from the longitudinal centerline **12** of the turbofan engine **10**. An angle **84** between the OGV reference line **74** and the radial reference line **82** is at least five (5) degrees and less than or equal to 45 degrees. In particular, the embodiment shown, the angle **84** is as least 15 degrees and less than or equal to 35 degrees. In such a manner, it will be appreciated that the outer junction **78** is located forward of the inner junction **76**.
(57) Including the outlet guide vane **52** being forward swept in the manner described hereinabove may allow for a shortening of the outer nacelle **50**. In particular, it will be appreciated that the outer nacelle **50** in the embodiment shown includes the thrust reverser assembly **54** and the closeout section **55** located aft of the thrust reverser assembly **54**. A length of the thrust reverser assembly **54** along the axial direction A may be difficult to shorten. Further, a relatively fixed length along the axial direction A may be required for the closeout section **55** for aerodynamic purposes. The inventors of the present disclosure found that the thrust reverser assembly **54** and closeout section **55** may not be moved forward of the tip **72** of the outlet guide vane **52**. Accordingly, in order to allow for the thrust reverser assembly **54** and closeout section **55** of the outer nacelle **50** to be moved forward, therefore allowing for an overall shorter outer nacelle **50** and turbofan engine **10**, the inventors of the present disclosure went against conventional turbofan engine design and provided the outlet guide vane **52** in the forward swept configuration.
(58) In particular, referring still also to FIG. **2**, it will be appreciated that the outer nacelle **50** includes a nacelle case **86**, with the tip **72** of the outlet guide vane **52** coupled to the nacelle case **86**. Further, the outer nacelle **50** includes an outer attachment groove **88** located aft of a trailing edge **90** of the outlet guide vane **52** at the tip **72**. For the embodiment shown, the outer attachment groove **88** is configured as part of the nacelle case **86**.
(59) As will be appreciated, in at least certain configurations, the outer attachment groove **88** must be positioned aft of the trailing edge **90** of the outlet guide vane **52** at the tip **72** (an aft-most connection between the outlet guide vane **52** and the nacelle case **86**), and the thrust reverser assembly **54** must be positioned aft of the outer attachment groove **88**. Accordingly, moving the aft-most connection between the outlet guide vane **52** and the nacelle case **86** forward, by virtue of including the outlet guide vane **52** in the forward slept configuration, may allow for moving the thrust reverser assembly **54** forward.
(60) Notably, the outer nacelle **50** further includes an aft nacelle liner **92** configured to attach to the outer attachment groove **88**. In particular, referring briefly also to FIG. **3**, providing a schematic view of the aft nacelle liner **92** and the outer attachment groove **88** along the axial direction A, the aft nacelle liner **92** includes a first section **94** and a second section **96**. The first section **94** and the second section **96** are each semicircular shape and together define a hinged connection **98** therebetween. Further, the first section **94** and the second section **96** each include a lip **100** (see FIG. **2**) which fits into the outer attachment groove **88**. The first section **94** and the second section **96** may be fitted over the outer attachment groove **88** and coupled at respective distal ends to attach

the aft nacelle liner **92** to the nacelle case **86**.

(61) Referring back particularly to FIG. **2**, it will be appreciated that the turbomachine **16** further includes an inner attachment groove **102**. The inner attachment groove **102** is configured as part of a frame of the turbomachine **16**, as will be discussed in more detail below. In particular, the outer casing **18** of the turbomachine **16** includes an aft outer casing liner **104**. The aft outer casing liner **104** may be coupled to the inner attachment groove **102** in a similar manner as the aft nacelle liner **92** is attached to the outer attachment groove **88**, described above. Accordingly, the aft outer casing liner **104** may be coupled to the inner attachment groove **102** of the turbomachine **16** in the same or similar manner as described above with reference to FIG. **3**.

(62) As will be appreciated, by virtue of the forward swept configuration of the outlet guide vane **52**, the inner attachment groove **102** is located aft of the outer attachment groove **88**. Further, in the exemplary embodiment of FIG. **2**, it will be appreciated that the outer attachment groove **88** and the inner attachment groove **102** are each configured as V-grooves (e.g., each groove having a "V" shape).

(63) It will be appreciated, however, that in other exemplary embodiments, the inner attachment groove **102**, the outer attachment groove **88**, or both may have other suitable configurations.

(64) Moreover, referring still to FIG. **2**, the base **70** of the outlet guide vane **52** is coupled to a frame of the turbomachine **16**. As will be appreciated, the turbomachine **16** includes a compressor forward frame **106** and an inter-compressor frame **108**. The compressor forward frame **106** is located forward of the LP compressor **22** along the working gas flowpath **37**. The inter-compressor frame **108** is located aft of the LP compressor **22** and forward of the HP compressor **24**.

(65) As will be appreciated from the view of FIG. **2**, the compressor forward frame **106** supports rotation of a fan shaft **110** (extending from the power gearbox **46** to the fan **38**) and further supports the LP shaft **36** through one or more bearings **112**. Notably, the fan shaft **110** is supported by the compressor forward frame **106** through a sump cone **114** (which may be an "A sump cone") extending from the compressor forward frame **106**.

(66) The base **70** of the outlet guide vane **52** is located aft of the compressor forward frame **106** and forward of the inter-compressor frame **108**. In such a manner, it will be appreciated that the outlet guide vane **52** defines the inner junction **76** between the outlet guide vane **52** and the turbomachine **16** at the leading edge **80** of the outlet guide vane **52**. The inner junction **76** is aligned with the LP compressor **22** along the longitudinal centerline **12**.

(67) In order to support the base **70** of the outlet guide vane **52** with such a configuration, the turbomachine **16** further includes a shell frame **116**, with the outlet guide vane **52** coupled to the shell frame **116**. The shell frame **116** is located between the compressor forward frame **106** and the inter-compressor frame **108**.

(68) In particular, for the embodiment depicted, the shell frame **116** extends between and is coupled to both the compressor forward frame **106** in the inter-compressor frame **108** to support the outlet guide vane **52**.

(69) Notably, since the base **70** of the outlet guide vane **52** is positioned aft of the compressor forward frame **106** for the embodiment shown, the compressor forward frame **106** may not be as stiff as it otherwise would be if the base **70** of the outlet guide vanes **52** were aligned with the compressor forward frame **106** along the axial direction A. Accordingly, for the embodiment shown, the turbomachine **16** includes a load reduction device **118** integrated into the sump cone **114** or into an attachment of the sump cone **114** (e.g., an attachment between the sump cone **114** and the compressor forward frame **106**). The load reduction device **118** may be a designated failure point within the sump cone **114** and/or within the attachment such that in the event of a failure condition (e.g., a blade-out condition of the fan **38**), excessive vibrations may cause the sump cone **114** to fail, insulating the compressor forward frame **106** from having to absorb such vibrations.

(70) In certain exemplary aspects, the load reduction device **118** may include a portion of the sump cone **114** having a reduced thickness, a portion of the sump cone **114** having perforations, etc.

(71) Referring still to FIG. **2**, the inventors found that inclusion of the outlet guide vane **52** configured in the forward swept orientation as described above may have an undesirable effect on the acoustics of the turbofan engine **10** at least in part due to a proximity of the tip **72** of the outlet guide vane **52** to the plurality of fan blades **40**. In particular, the inventors found that such configuration may generate acoustic waves, further that such a configuration may direct at least a portion of said acoustic waves inwardly along the radial direction R from the outlet guide vane **52**.
(72) However, the inventors further found that the undesirable effect on the acoustics of the turbofan engine **10** may be addressed by further including an acoustic treatment **120** attached to or integrated with the outer casing **18** at a location aligned with the outlet guide vanes **52** along the longitudinal centerline **12**.
(73) It will be appreciated that as used herein, the term "aligned with the outlet guide vane **52** along the longitudinal centerline **12**" refers to any position along the longitudinal centerline **12** between a forward-most location of the outlet guide vane **52** and an aft-most location of the outlet guide vane **52**.
(74) In particular, in the embodiment shown, the outlet guide vane **52** defines the inner junction **76** with the outer casing **18** of the turbomachine **16** at the leading edge **80** of the outlet guide. The acoustic treatment **120** is located at the inner junction **76** (e.g., aligned with the inner junction **76** along the longitudinal centerline **12**), forward of the inner junction **76**, or both. More specifically, it will be appreciated that the turbomachine **16** defines a distance along the axial direction A from the inlet **20** of the turbomachine **16** to the inner junction **76**. The acoustic treatment **120** extends along the axial direction A at least 50% of the distance, such as at least 75% of the distance. More specifically, still, from embodiment depicted, the acoustic treatment **120** extends 100% of the distance along the axial direction A from the inlet to the inner junction **76**.
(75) In the embodiment depicted, the acoustic treatment **120** includes three acoustic treatments **120** spaced along the axial direction A between the inner junction **76** and the inlet **20**. The turbofan engine **10** further includes an additional acoustic treatment **120** at least partially aft of the inner junction **76**.
(76) Further, for the embodiment depicted, in order to further attenuate noise generated as a result of the forward swept configuration of the outlet guide vane **52**, the turbofan engine **10** further includes an acoustic treatment **120** integrated into the outlet guide vane **52** on a pressure side, on a suction side, or both. In particular, for the embodiment shown, the acoustic treatment **120** is integrated into the pressure side.
(77) Referring now to FIG. **4**, it will be appreciated that the acoustic treatment **120** extends along a circumferential direction C, and more specifically extends substantially continuously along the circumferential direction C. In particular, FIG. **4** provides a schematic view of the outer casing **18** of the turbomachine **16** of FIG. **2** at a location between the inlet **20** and the inner junction **76**. In the embodiment shown, the acoustic treatment **120** extends 360 degrees in the circumferential direction C.
(78) The acoustic treatment **120** incorporated may have any suitable configuration, or any suitable combination of configurations.
(79) In at least certain exemplary embodiments, the acoustic treatment **120** may include a perforated sheet **122** with a hollow body **124**. In particular, referring now to FIG. **5**, a schematic, cross-sectional view is provided of an acoustic treatment **120** in accordance with an exemplary aspect of the present disclosure. As shown in the embodiment of FIG. **5**, the exemplary acoustic treatment **120** includes the perforated sheet **122** and the hollow body **124**. The hollow body **124** includes a liner **128** defining an interior void adjacent to the perforated sheet **122**. The perforated sheet **122** defines a plurality of openings **126** allowing an external environment to communicate with the interior void of the hollow body **124**. Acoustic waves may enter the hollow body **124** through the plurality of openings **126**, allowing for an attenuation of the noise generated by virtue of the orientation of the outlet guide vanes **52** in the forward swept arrangement.

(80) The perforated sheet **122** is coupled to the liner **128** through a plurality of extensions **130** extending from the perforated sheet **122** to the liner **128**. In certain exemplary embodiments, the acoustic treatment **120** may further include additional structures to increase noise attenuation achieved by the acoustic treatment **120** at desired frequencies. The additional structures may be walls or other extensions **130** (depicted in phantom) extending from the perforated sheet **122**, extending from the liner **128**, or both; may be perforations in the walls or extensions **130**; may be additional or alternative walls or extensions **130**; etc.
(81) Referring now to FIG. **6**, a schematic, a top view is provided of a perforated sheet **122** in accordance with an exemplary aspect of the present disclosure. The perforated sheet **122** may include a plurality of openings **126** spaced in a uniform manner.
(82) Referring now to FIG. **7**, a schematic, a top view is provided of a perforated sheet **122** accordance with another exemplary aspect of the present disclosure. As will be appreciated from the view of FIG. **7**, the plurality of openings **126** of the perforated sheet **122** may define a noncircular shape, such as an elongated or ovular shape.
(83) Referring now to FIG. **8**, a schematic, a top view is provided of a perforated sheet **122** in accordance with yet another exemplary embodiment of the present disclosure. As will be appreciated from the view of FIG. **8**, the plurality of openings **126** may define a nonuniform size and a nonuniform spacing. Such a configuration may, e.g., allow for the acoustic treatment **120** to target noise at various frequencies.
(84) Referring briefly back to FIG. **1**, it will be appreciated that although a single outlet guide vane **52** is described hereinabove with reference to FIGS. **1** through **7**, the turbofan engine **10** further includes a plurality of outlet guide vanes **52**. Each of the plurality of outlet guide vanes **52** may be oriented in substantially the same manner as the exemplary outlet guide vanes **52** described above. Accordingly, it will be appreciated that each outlet guide vane **52** of the plurality of outlet guide vanes **52** may define a base **70** and a tip **72** and be forward swept from the base **70** to the tip **72** in the same manner as the outlet guide vane **52** described above. The plurality of outlet guide vanes **52** may be spaced along the circumferential direction C, as is depicted schematically, e.g., in FIG. **4**.
(85) Moreover, as described above, orienting the plurality of outlet guide vanes **52** in a forward swept configuration may allow for movement of, e.g., the thrust reverser assembly **54** and closeout section **55** of the outer nacelle **50** forward. In order to further reduce an overall length of the outer nacelle **50** and turbofan engine **10**, the turbofan engine **10** in the embodiment shown further includes a relatively short inlet section. In particular, it will be appreciated that the outer nacelle **50** defines an inlet length L. The inlet length L refers to a distance along the axial direction A between a leading edge **132** of the outer nacelle **50** and a leading edge of the fan blades **40** where the fan blades **40** meet the hub **48**. In the embodiment shown, the turbofan engine **10** defines a ratio of the inlet length L to the fan diameter D equal to or less than 0.5.
(86) The inclusion of the plurality of outlet guide vanes **52** oriented in the forward swept configuration, and further including acoustic treatment(s) **120** in accordance with one or more exemplary aspects of the present disclosure, may allow for the turbofan engine **10** to achieve the desired shorter length, while addressing any acoustic effects resulting from such configuration.
(87) Referring now to an additional and/or alternative exemplary configuration of the present disclosure, it will be appreciated that the demand for more aerodynamically efficient and compact turbofan engines remains strong. Prior design of turbofan engines has favored the placement of an accessory gearbox ("AGB") within an outer nacelle of the turbofan engines, e.g., due to the conventionally more favorable environmental conditions and a goal of saving space within an outer casing of a turbomachine of the turbofan engines. This arrangement, in combination with traditionally used radially-oriented or aft-swept outlet guide vanes (OGVs), may result in a longer overall turbofan engine, including a longer outer nacelle. Though this design may provide suitable environmental conditions for the AGB, the inventors of the present disclosure have found that it

may result in detrimental aerodynamic lines and overall engine efficiency.
(88) In particular, contrary to previous turbofan engine design, the inventors have found that by including forward-swept OGVs, a length of the outer nacelle may be significantly shortened. As part of this modification, however, the inventors found that such a reduction would pose challenges for the placement of the AGB in the outer nacelle, potentially resulting in aerodynamic compromises or requiring an otherwise unnecessary elongation of the outer nacelle. Accordingly, the inventors found that by positioning the AGB, or at least part of it, inside the outer casing of the turbomachine, the above issues with the outer nacelle may be avoided, resulting in a balanced approach to maintain turbofan engine compactness without compromising desired aerodynamic and efficiency properties.
(89) Referring now to FIG. **9**, a schematic view of a turbofan engine **10** in accordance with another exemplary embodiment of the present disclosure is provided. The exemplary turbofan engine **10** of FIG. **9** may be configured in a similar manner as the exemplary turbofan engine **10** described above with reference to FIGS. **1** through **8**.
(90) For example, the exemplary turbofan engine **10** of FIG. **9** generally includes an outer nacelle **50** and a turbomachine **16**. The outer nacelle **50** includes a nacelle case **86**, which in the embodiment depicted, surrounds and obstructs from view, a fan section **14** having a fan **38** and a plurality of outlet guide vanes **52**. The plurality of outlet guide vanes **52** may be configured in substantially the same manner as the exemplary outlet guide vanes **52** of the turbofan engine **10** described above. Accordingly, the plurality of outlet guide vanes **52** may each define a base **70** and a tip **72** and be forward swept from the base **70** to the tip **72**.
(91) Further, the turbomachine **16** includes an outer casing **18** and an accessory gearbox **150**. For the embodiment shown, the accessory gearbox **150** is positioned at least partially inward of the outer casing **18** of the turbomachine **16** along the radial direction R. In such a manner, it will be appreciated that the accessory gearbox **150** may be at least partially confined within the outer casing **18**.
(92) In certain embodiments, the accessory gearbox **150** may be mounted or operably coupled to an engine core **152** of the turbomachine **16** with a hinge mount. That is, when one or more clasps or fasteners is removed, the accessory gearbox **150** can swing away from at least part of the engine core **152** to which a remainder of the accessory gearbox **150** remains operably coupled to. The accessory gearbox **150** is operably coupled to the engine core **152** in a location and orientation so as to provide easy access to the accessory gearbox **150** and other component such as, but not limited to, fuel lines, electrical cables, electrical connectors, oil tubes, sight glasses, and fill ports.
(93) The accessory gearbox **150** defines an AGB axis **154**. In the illustrated example, the AGB axis **154** is parallel to the turbine engine axis of rotation (longitudinal centerline **12**) when the accessory gearbox **150** is fully installed and in use within the turbine engine **10**. It is contemplated, however, that the AGB axis **154** and the turbine engine axis of rotation can be at any suitable angle and need not be parallel.
(94) More specifically, for the exemplary embodiment depicted, the accessory gearbox **150** includes a first portion **156** and a second portion **158**. The first portion **156** is positioned at least partially within the outer casing **18** of the turbomachine **16** and the second portion **158** is positioned at least partially outside of the outer casing **18**, e.g., along a radial direction R of the turbofan engine **10**. In particular, for the embodiment depicted, the first portion **156** is positioned entirely within the outer casing **18**. In particular, in the embodiment shown, the first portion **156** of the accessory gearbox **150** can straddle the engine core **152** within the outer casing **18**.
(95) More specifically, still, from embodiment depicted, the turbofan engine **10** includes one or more struts **160** extending between the turbomachine **16** and the outer nacelle **50**. In particular, the turbofan engine **10** depicted includes an upper strut **160**A and a lower strut **160**B. The first portion **156** of the accessory gearbox **150** is positioned inward of the outer casing **18** of the turbomachine **16** along the radial direction R and the second portion **158** of the accessory gearbox **150** extends

into the lower strut **160**B of the turbofan engine **10** for the embodiment depicted.

(96) In such a manner, it will be appreciated that for the embodiment depicted, the second portion **158** of the accessory gearbox **150** is located (at least partially) between the outer casing **18** and the outer nacelle **50**. Referring still to FIG. **9**, the second portion **158** of the accessory gearbox **150** is illustrated as being perpendicular to the AGB axis **154**.

(97) However, in other embodiments, it is further contemplated that the second portion **158** and the AGB axis **154** can be at any relative angle.

(98) Referring now to FIG. **10**, FIG. **10** further illustrates the accessory gearbox **150** provided with the engine core **152**. The body of the outer nacelle **50** (FIG. **9**), outer casing **18** (FIG. **9**), and internal portions of the engine core **152** have not been illustrated for clarity. A set of forks illustrated as a first fork **162** and a second fork **164** can be included in the accessory gearbox **150**. As used herein, the term "fork" is a Y-shaped object having an upright or base from which two arms branch off in different directions.

(99) In the illustrated example, the first fork **162** includes a first base portion **166** defined by a part of the second portion **158** of the accessory gearbox **150**. A first arm **168** and a second arm **170** (not shown as it wraps behind the engine core **152**) extend from the first base portion **166**. The first arm **168** and the second arm **170** are part of the first portion **156** of the accessory gearbox **150** that can straddle the engine core **152**. That is, the first base portion **166** extends, forks, splits, branches, or otherwise couples to the first arm **168** and the second arm **170**, where the first arm **168** and the second arm **170** form a V-shape, U-shape, or the like, in order to cradle, straddle, or otherwise partially circumscribe the engine core **152**.

(100) Similarly, the second fork **164** can be defined by a second base portion that forks, splits, or otherwise couples to a first arm **172** and a second arm (not shown as it wraps behind the engine core **152**) that straddle the engine core **152**. It is contemplated that any number of one or more forks can be included in the accessory gearbox **150**.

(101) The first fork **162** and the second fork **164** are illustrated as being spaced along the AGB axis **154**. The first fork **162** and the second fork **164** can be coupled via one or more of a drive shaft, gearbox, hydraulic drive, or the like. The drive shaft or a hydraulic drive can include one or more casings **174** positioned between the first fork **162** and the second fork **164**. It is contemplated that only a single fork can be included or that additional sections, casing, drive shafts, or the like, can be included to increase the number of interfaces or forks.

(102) The accessory gearbox **150** can couple to any number of interfaces, illustrated, by way of non-limiting example as interfaces **174***a*, **174***b*, **174***c*. Additionally, it is contemplated that the accessory gearbox **150** can couple to any number of components or systems illustrated, by way of non-limiting example as systems **133***a*, **133***b*, **133***c*. Systems powered by the accessory gearbox **150** can be located at axially or radially spaced locations relative to the accessory gearbox **150**. By way of example, systems or interfaces can include, but are not limited to, any one or more of an output shaft, fuel pump, transfer gearbox, lubrication pump, air compressor, scavenge pump, electrical generator, fuel control, fuel pump, permanent magnet alternator, lubrication pump, or hydraulic pump.

(103) Referring now to FIG. **11**, FIG. **11** is a cross-section further illustrating the accessory gearbox **150**. The cross-section is a schematic cross-section generally taken at an axial location of the first fork **162** (FIG. **10**). The engine core **152** is schematically illustrated along with the turbine engine longitudinal centerline **12**. A C-shaped or arc cross-section can be defined by the first arm **168** and the second arm **170** and straddle the engine core **152**. While no spacing or coupling components are illustrated between the engine core **152** and the accessory gearbox **150**, it will be understood that any suitable spacing and components can be included. The first arm **168** extends from the AGB axis **154** to a first terminating point or a first distal surface **176**. The first distal surface **176** can have a first end point **178** defined as the point on the first distal surface **176** that is radially farthest from the turbine engine longitudinal centerline **12**. A first distance **180** can be measured radially

from the AGB axis **154** to the first end point **178**.

(104) The first arm **168** can include a first plane **182**. The two dimensions defining the first plane **182** are illustrated by way of example as a first dimension that is generally perpendicular to the first distal surface **176** and a second dimension that is into/out of the page. As used herein, the term “generally perpendicular” defines an angle between two objects that is between 80 degrees and 100 degrees (inclusive of the endpoints). Additionally, or alternatively, the first plane **182** can be defined by a plane defined by at least an obtuse angle extending from the yaw or pitch axis (e.g., from into/out of page, or top to bottom respectively in FIG. **9**). In FIG. **11**, the obtuse angle is formed with a yaw-roll plane. In other examples, the first plane **182** can be defined by rotation around more than one of these three mutually orthogonal axes. It is further contemplated that the first plane **182** can form any angle in either dimension with the first distal surface **176**.

(105) A primary set of interfaces **184***a*, **184***b* can be operably coupled to or defined by a portion of the first arm **168**. The primary set of interfaces **184***a*, **184***b* can extend along or be defined by the first plane **182**. While illustrated as two interfaces **184***a*, **184***b* that lie on the first plane **182**, any number of interfaces in the first plane **182**, including one, are contemplated.

(106) The first arm **168** can define a first arm axis of rotation **186**. It is contemplated that the rotation of an output shaft or other portion of the first arm **168** about the first arm axis of rotation **186** can be considered an interface of the accessory gearbox **150**. An interface as defined herein can be an input to the accessory gearbox **150** or an output from the accessory gearbox **150**. It is further contemplated that the first arm **168** can define any number of axes of rotations.

(107) The first arm **168** can sweep a first arclength **188** from the AGB axis **154** to the first distal surface **176**. The first arm **168** can cover, straddle, or otherwise wrap around between 5% and 50% of the engine core **152**. It is contemplated that the length of the first arclength **188** is between 1% and 60% the circumference of the engine core **152**.

(108) The second arm **170** extends from the plane of the AGB axis **154** to a second terminating point or a second distal surface **190**. The second distal surface **190** can have a second end point **192** defined as the point on the second distal surface **190** that is radially farthest from the turbine engine longitudinal centerline **12**. A second distance **194** can be measured radially from the AGB axis **154** to the second end point **192**. While illustrated as equal, the first distance **180** can be greater than or less than the second distance **194**. That is, the second portion **158** of the accessory gearbox **150** is spaced non-equidistant between the first end point **178** and the second end point **192**, where the first end point **178** and the second end point **192** are defined by the arc cross-section.

(109) The second arm **170** can include a second plane **196**. The two dimensions defining the second plane **196** are illustrated by way of example as a first dimension that is generally perpendicular to the second distal surface **190** and a second dimension that is into/out of the page. Similar to the first plane **182**, the second plane **196** can be defined by a plane defined by at least an obtuse angle extending from the yaw or pitch axis (i.e., from into/out of page, or top to bottom respectively in FIG. **9**). It is further contemplated that the second plane **196** can form any angle in either dimension with the second distal surface **190**.

(110) A secondary set of interfaces **198***a*, **198***b* can be operably coupled to or defined by a portion of the second arm **170**. The secondary set of interfaces **198***a*, **198***b* can be defined by the second plane **196**. While illustrated as two interfaces **198***a*, **198***b* that lie on or extend along the second plane **196**, any number of interfaces in the second plane **196**, including one, are contemplated.

(111) The second arm **170** can define a second arm axis of rotation **200**. It is contemplated that the rotation of an output shaft or other portion of the second arm **170** about the second arm axis of rotation **200** can be considered an interface of the accessory gearbox **150**. It is further contemplated that the second arm **170** can define any number of axes of rotations.

(112) The second arm **170** can sweep a second arclength **202** from the AGB axis **154** to the second distal surface **190**. The second arm **170** can cover or otherwise wrap around between 5% and 50% of the engine core **152**. It is contemplated that the length of the second arclength **202** is between

1% and 60% the circumference of the engine core **152**. While illustrated as equal, the second arclength **202** can be less than or greater than the first arclength **188**.
(113) A third plane **204** can define a tertiary set of interfaces **206***a*, **206***b* included in the second portion **158** of the accessory gearbox **150**. As illustrated, by way of non-limiting example, the second portion **158** of the accessory gearbox **150** can be generally perpendicular to the AGB axis **154**. The tertiary set of interfaces **206***a*, **206***b* can extend along the third plane **204**. It is contemplated that one or more of the tertiary set of interfaces **206***a*, **206***b* extends past an inner surface (along the radial direction R) of the outer nacelle **50**. That is, the accessory gearbox **150** can include the primary set of interfaces **184***a*, **184***b* defined by the first plane **182**, the secondary set of interfaces **198***a*, **198***b* defined by the second plane **196**, and the tertiary set of interfaces **206***a*, **206***b* defined by the third plane **204**. Interfaces can include, but are not limited to systems, components, or other engine elements that receive energy from the rotation of the accessory gearbox **150** about the AGB axis **154**. It is contemplated that the interfaces **206***a*, **206***b*, **184***a*, **184***b*, **198***a*, and **198***b* can correspond to an interface to service one or more of, or any of an output shaft, fuel pump, transfer gearbox, lubrication pump, air compressor, scavenge pump, electrical generator, fuel control, fuel pump, permanent magnet alternator, lubrication pump, or hydraulic pump. It is further contemplated that the second portion **158** of the accessory gearbox **150** can define any number of axes of rotations, and in the embodiment depicted the second portion **158** defines an axis of rotation **159**.
(114) The third plane **204**, similar to the first plane **182** and the second plane **196**, includes two dimensions illustrated by way of example as a first dimension that is generally perpendicular to the AGB axis **154**, extending radially outward from the longitudinal centerline **12**, and a second dimension that is into/out of the page. Similar to the first plane **182** and the second plane **196**, the third plane **204** can be defined by a plane defined by at least an obtuse angle extending from the yaw or pitch axis (i.e., from into/out of page, or top to bottom respectively in FIG. **9**). It is further contemplated that the third plane **204** can form any angle in either dimension with the AGB axis **154**.
(115) As shown, by way of example, each of the respective pair of interfaces **206***a*, **206***b*, **184***a*, **184***b*, **198***a*, and **198***b* or axes of rotations **186**, **200**, **159** associated with the respective planes **182**, **196**, **204** are generally perpendicular to each other. However, it is contemplated that the angles between the interfaces **206***a*, **206***b*, **184***a*, **184***b*, **198***a*, and **198***b* or the axes of rotations **186**, **200**, **159** associated with the respective planes **182**, **196**, **204** can be any angle, including between 50 degrees to 100 degrees.
(116) A first angle **208** can be defined from the third plane **204** to the first plane **182** by a clockwise rotation. As illustrated, the first angle **208** can be an acute angle, however any angle between, but not including, zero degrees and 180 degrees is contemplated. A second angle **210** can be from the third plane **204** to the second plane **196** by a counterclockwise rotation. As illustrated, the first angle **208** can be an acute angle, however any angle between, but not including, zero degrees and 180 degrees is contemplated. While illustrated as equal, it is contemplated that the first angle **208** can be greater than or less than the second angle **210**.
(117) It is contemplated that any number of additional planes defining interfaces or axes of rotation can extend from or be defined by the first portion **156** or the second portion **158**. It is also contemplated that the additional planes of interfaces or axes of rotation can extend from or be defined by one or more portions of the first plane **182**, the second plane **196**, the third plane **204**, the first arm axis of rotation **186**, the second arm axis of rotation **200**, the AGB axis **154**, the first arclength **188**, or the second arclength **202**.
(118) In operation, the accessory gearbox **150** is operably coupled to one or more components of the engine core **152**. That is, one or more components of the engine core **152** provides or otherwise communicates energy to the accessory gearbox **150**. By way of non-limiting example, the accessory gearbox **150** can be powered by energy provided by a drive shaft located in the engine

core **152** along the turbine engine longitudinal centerline **12** or the LP shaft **36** (see FIG. **1**). Additionally, or alternatively, the accessory gearbox **150** can be electrically driven using electrical power generated by the rotation of the engine core **152** or a storage device for electrical energy.
(119) The accessory gearbox **150**, when powered, rotates one or more components about the AGB axis **154**. The rotation about the AGB axis **154** can then provide energy in the form of rotational energy or electro-magnetic energy to at least one of the interfaces from the primary set of interfaces **184***a*, **184***b*, the secondary set of interfaces **198***a*, **198***b*, or the tertiary set of interfaces **206***a*, **206***b*. The primary set of interfaces **184***a*, **184***b*, the secondary set of interfaces **198***a*, **198***b*, and the tertiary set of interfaces **206***a*, **206***b* are located in the first plane **182**, the second plane **196**, and the third plane **204**, respectively. The first plane **182**, the second plane **196**, and the third plane **204** are three distinct planes that can be parallel or intersect.
(120) The transfer of energy from the accessory gearbox **150** to the primary set of interfaces **184***a*, **184***b*, the secondary set of interfaces **198***a*, **198***b*, and the tertiary set of interfaces **206***a*, **206***b* can be from the rotation of a bevel gear arrangement. That is, one or more of the primary set of interfaces **184***a*, **184***b*, the secondary set of interfaces **198***a*, **198***b*, or the tertiary set of interfaces **206***a*, **206***b* can be driven using a system of bevel gears having a shared drive shaft. It is further contemplated that the accessory gearbox **150** can include multiple motors to provide power to hydraulic or electrically driven interfaces.
(121) The tertiary set of interfaces **206***a*, **206***b* coupled to or defined by the second portion **158** of the accessory gearbox **150** allow the first portion **156** of the accessory gearbox **150** to be smaller. That is, having the second portion **158** of the accessory gearbox **150** allows the distance between the outer casing **18** and the engine core **152** to decrease. The smaller distance between the outer casing **18** and the engine core **152** can increase the aerodynamics of a bypass airflow duct **56** (see FIG. **9**) by decreasing a drag on an airflow therethrough. Additionally, or alternatively, the smaller first portion **156** of the accessory gearbox **150** can provide room for additional components.
(122) The accessory gearbox **150** having the first portion **156** between the engine core **152** and the outer casing **18** and the second portion **158** between the outer casing **18** and the outer nacelle **50** allows for the turbofan engine **10** to include, for example, a slim line nacelle fan cowl and/or advanced thrust reverser assemblies **54** (FIG. **1**). In particular, inclusion of the exemplary accessory gearbox **150** may allow for the benefits associated with inclusion of the outlet guide vanes **52** (FIG. **1**) in the forward swept configuration to be realized (e.g., a shorter outer nacelle **50**, as increased length may not be needed to accommodate a thickness of an entirety of an accessory gearbox located in the outer nacelle **50**).
(123) It will be appreciated, however, that the exemplary turbofan engine **10** and accessory gearbox **150** described above with reference to FIG. **9** through **11** is provided by way of example only. In other exemplary embodiments, the turbofan engine **10** and accessory gearbox **150** may have any other suitable configuration.
(124) For example, referring briefly to FIG. **12**, a turbofan engine **10** in accordance with another exemplary embodiment of the present disclosure is provided. The exemplary turbofan engine **10** of FIG. **12** is configured in substantially the same manner as the exemplary turbofan engine **10** described above with reference to FIGS. **9** through **11**. For example, the exemplary turbofan engine **10** of FIG. **12** includes an accessory gearbox **150**. However, for the exemplary embodiment of FIG. **12**, the accessory gearbox **150** is positioned entirely within an outer casing **18** of a turbomachine **16** of the turbofan engine **10**.
(125) Further by way of example, referring now to FIG. **13**, a turbofan engine **10** in accordance with another exemplary embodiment of the present disclosure is provided. The exemplary turbofan engine **10** of FIG. **13** is also configured in a similar manner as the exemplary turbofan engine **10** described above with reference to FIGS. **9** through **11**. For example, the exemplary turbofan engine **10** of FIG. **13** includes an accessory gearbox **150**. The accessory gearbox **150** includes a first portion **156** and a second portion **158**. The first portion **156** is positioned at least partially within an

outer casing **18** of a turbomachine **16** of the turbofan engine **10**. In particular, for the embodiment shown the first portion **156** is positioned entirely within the outer casing **18** of the turbomachine **16**.
(126) However, for the embodiment depicted, the second portion **158** of the accessory gearbox **150** is positioned within an outer nacelle **50** of the turbofan engine **10**. Notably, the exemplary accessory gearbox **150** includes a drive assembly **212** extending from the first portion **156** to the second portion **158**, mechanically coupling the first portion **156** and the second portion **158** such that the first portion **156** may share rotational power with the second portion **158** (and vice versa). The drive assembly **212** may include a mechanical drive assembly (e.g., a combination of shafts, gears, etc.), a hydraulic drive assembly, or a combination thereof. In particular, for the embodiment depicted, the drive assembly **212** extends into a strut **160** (and more specifically a lower strut **160**B) of the engine **100**.
(127) The inclusion of accessory gearbox **150** in accordance with one or more the exemplary aspects discussed above in combination with a turbofan engine **10** in accordance with one or more exemplary aspects discussed herein, may allow for an overall shorter turbofan engine **10**, improving, e.g., aerodynamics and efficiency for the turbofan engine **10**. In particular, by mounting at least a portion of the accessory gearbox **150** within an outer casing **18** of a turbomachine **16** of the turbofan engine **10**, contrary to conventional turbofan engine **10** design principles suggesting that the accessory gearbox **150** should be positioned in the outer nacelle **50** for, e.g., more desirable environmental conditions and to save under cowl space on the turbomachine **16**, benefits associated with shortening an axial length of the outer nacelle **50** achieved through inclusion of forward swept outlet guide vanes **52** may be realized. In particular, positioning of the accessory gearbox **150** in accordance with the present disclosure may allow for a thinner outer nacelle **50**. If the entirety of the accessory gearbox **150** were instead positioned in the outer nacelle **50**, the outer nacelle **50** may require include a bulge (e.g., a local increased thickness) that would require additional length to be smoothened out from an aerodynamics standpoint.
(128) Referring now to an additional and/or alternative exemplary configuration of the present disclosure, it will be appreciated that the demand for more aerodynamically efficient and compact turbofan engines remains strong. Turbofan engines are generally designed to have a lower Fan Pressure Ratio (FPR) to produce efficient thrust. The low FPR may result in a reduction in fan flow separation.
(129) One way to achieve a low FPR is to incorporate a reduction gearbox to allow the fan to rotate slower than a drive turbine (e.g., an LP turbine). Such may introduce weight and complexity. Alternatively, the fan may be directly driven by the drive turbine. However, in order to achieve a low FPR, the drive turbine is constrained in how fast it can rotate, which may result in a less efficient drive turbine.
(130) In particular, challenging this previous thinking, the inventors have found that introducing inlet pre-swirl features upstream of the fan blades can address the efficiency and separation challenges associated with high-speed fans. This approach not only enables improved fan performance, but also facilitates the integration of a high-speed booster. Such a booster, in turn, paves the way for a high-speed drive turbine (e.g., the LP turbine) with potential modifications like a reduced rotor stage count and/or an increased radius, which may result in a more compact turbomachine.
(131) Further, incorporating forward-swept OGVs to this arrangement may further allow for a reduction in a length of an outer nacelle of the turbofan engine. Thus, by integrating the forward-swept OGVs with the inlet pre-swirl features, the inventors have found that an overall length of the turbofan engine may be reduced, while also improving aerodynamic performance and engine efficiency.
(132) Notably, the increased speed of the fan may further bolster conventional thinking that the outlet guide vanes should be further spaced from the fan blades of the fan (discouraging incorporation of forward swept OGVs in a turbofan engine with a high speed fan). However,

utilizing one or more of the configurations discussed hereinabove, the inventors of the present disclosure have overcome these concerns.
(133) Referring now to FIG. **14**, a turbofan engine **10** in accordance with another exemplary embodiment of the present disclosure is provided. The exemplary turbofan engine **10** of FIG. **14** may be configured in a similar manner as one or more of the exemplary turbofan engines **10** described above. For example, the exemplary turbofan engine **10** of FIG. **14** generally includes a fan section **14** having a fan **38** with a plurality of fan blades **40**, a turbomachine **16** drivingly coupled to the fan **38** and having an outer casing **18**, and an outer nacelle **50** surrounding the fan **38** at least a portion of the turbomachine **16**. Additionally, the exemplary turbofan engine **10** of FIG. **14** further includes an outlet guide vane **52** extending between the turbomachine **16** and the outer nacelle **50** at a location downstream of the fan blades **40** of the fan **38**. The outlet guide vane **52** defines a base **70** and a tip **72** and is forward swept from the base **70** to the tip **72**. More specifically, the turbofan engine **10** includes a plurality of outlet guide vanes **52** configured in such a manner.
(134) However, it will be appreciated that turbofan engine **10** is configured as a "direct-drive" turbofan engine **10**. More specifically, for the embodiment shown, the fan **38** of the fan section **14** is driven by an LP turbine **30** of the turbomachine **16** directly (i.e., the LP turbine **30** rotates at the same rotational speed as the fan **38**). In such a manner, it will be appreciated that the fan **38**, and an LP compressor **22** rotatable with the fan **38**, are configured to rotate at a relatively high rotational speed. In order to reduce an amount of flow separation, e.g., at outer tips of the plurality of fan blades **40** of the fan **38**, the turbofan engine **10** of the present disclosure also provides a pre-swirl feature attached to or integrated into the outer nacelle **50** at a location upstream of the plurality of fan blades **40** of the fan **38**.
(135) More specifically, for the exemplary embodiment of FIG. **14**, the inlet pre-swirl feature is one of a plurality of inlet pre-swirl features attached to or integrated into the outer nacelle **50** at a location upstream of the plurality of fan blades **40**, and the plurality of inlet pre-swirl features are each configured as a part span inlet guide vane **300**, as described in more detail below.
(136) Briefly, it will be appreciated that by configuring the turbofan engine **10** in such a manner, a benefit associated with including the plurality of outlet guide vanes **52** oriented in a forward swept configuration may be realized. In particular, by having the fan **38** configured in a direct drive configuration with the LP turbine **30**, with the plurality of inlet pre-swirl features, the fan **38** and LP turbine **30** may rotate more quickly, allowing for the LP turbine **30** to have a shorter axial length (e.g., a lower number of stages of LP turbine rotor blades and/or a higher radius). In such a manner, an overall length for the turbomachine **16** may be reduced, allowing for a benefit associated with having a reduced length outer nacelle **50** (achieved through the forward swept outlet guide vanes **52**), to be realized.
(137) Notably, referring now also to FIG. **15**, providing a close-up, schematic view of the fan section **14** and forward end of the turbomachine **16** of the exemplary turbofan engine **10** of FIG. **14**, for the embodiment depicted the turbofan engine **10** further includes one or more acoustic treatments **120** coupled to or integrated into the part span inlet guide vanes **300**, an inner wall **51** of the outer nacelle **50**, or both. In particular, for the embodiment depicted, the turbofan engine **10** includes a plurality of acoustic treatments **120** coupled to or integrated into the part span inlet guide vanes **300** (e.g., on a pressure or suction side), the inner wall **51** of the outer nacelle **50** forward of the part span inlet guide vanes **300**, and the inner wall **51** of the outer nacelle **50** aft of the part span inlet guide vanes **300** (and upstream of the plurality of fan blades **40**). Such a configuration may further assist with attenuation of noise associated with the forward swept outlet guide vanes **52**.
(138) Referring still to FIG. **15**, in the embodiment depicted, the plurality of part span inlet guide vanes **300** are each cantilevered from the outer nacelle **50** (such as from the inner wall **51** of the outer nacelle **50**) at a location forward of the plurality of fan blades **40** of the fan **38** along the axial direction A and aft of an inlet **60** of the outer nacelle **50**. More specifically, each of the plurality of

part span inlet guide vanes **300** defines an outer end **302** along the radial direction R, and are attached to/connected to the outer nacelle **50** at the radially outer end **302** through a suitable connection means (not shown). For example, each of the plurality of part span inlet guide vanes **300** may be bolted to the inner wall **51** of the outer nacelle **50** at the outer end **302**, welded to the inner wall **51** of the outer nacelle **50** at the outer end **302**, or attached to the outer nacelle **50** in any other suitable manner at the outer end **302**.
(139) Further, for the embodiment depicted, the plurality of part span inlet guide vanes **300** extend generally along the radial direction R from the outer end **302** to an inner end **304** (i.e., an inner end **304** along the radial direction R). Moreover, as will be appreciated, for the embodiment depicted, each of the plurality of part span inlet guide vanes **300** is unconnected with an adjacent part span inlet guide vane **300** at the respective inner ends **304** (i.e., adjacent part span inlet guide vanes **300** do not contact one another at the radially inner ends **304**, and do not include any intermediate connection members at the radially inner ends **304**, such as a connection ring, strut, etc.). More specifically, for the embodiment depicted, each part span inlet guide vane **300** is completely supported by a connection to the outer nacelle **50** at the respective outer end **302** (and not through any structure extending, e.g., between adjacent part span inlet guide vanes **300** at a location inward of the outer end **302** along the radial direction R). Such a configuration may reduce an amount of turbulence generated by the part span inlet guide vanes **300**.
(140) Moreover, as depicted, each of the plurality of part span inlet guide vanes **300** does not extend completely between the outer nacelle **50** and, e.g., the hub **48** of the turbofan engine **10**. More specifically, for the embodiment depicted, each of the plurality of inlet guide vane defines an inlet guide vane ("IGV") span **306** along the radial direction R, and further each of the plurality of part span inlet guide vanes **300** also defines a leading edge **308** and a trailing edge **310**. The IGV span **306** refers to a measure along the radial direction R between the outer end **302** and the inner end **304** of the part span inlet guide vane **300** at the leading edge **308** of the part span inlet guide vane **300**. Similarly, the plurality of fan blades **40** of the fan **38** define a fan blade span **312** along the radial direction R. More specifically, each of the plurality of fan blades **40** of the fan **38** also defines a leading edge **314** and a trailing edge **316**, and the fan blade span **312** refers to a measure along the radial direction R between a radially outer tip and a base of the fan blade **40** at the leading edge **314** of the respective fan blade **40**.
(141) For the embodiment depicted, the IGV span **306** is at least about five percent of the fan blade span **312** and less than or equal to about fifty-five percent of the fan blade span **312**. For example, in certain exemplary embodiments, the IGV span **306** may be between about fifteen percent of the fan blade span **312** and about forty-five percent of the fan blade span **312**, such as between about thirty percent of the fan blade span **312** and about forty percent of the fan blade span **312**.
(142) Reference will now also be made to FIG. **16**, providing an axial view of the inlet **60** to the turbofan engine **10** of FIGS. **14** and **15**. As will be appreciated, for the embodiment depicted, the plurality of part span inlet guide vanes **300** of the turbofan engine **10** includes a relatively large number of part span inlet guide vanes **300**. More specifically, for the embodiment depicted, the plurality of part span inlet guide vanes **300** includes between about ten part span inlet guide vanes **300** and about fifty part span inlet guide vanes **300**. More specifically, for the embodiment depicted, the plurality of part span inlet guide vanes **300** includes between about twenty part span inlet guide vanes **300** and about forty-five part span inlet guide vanes **300**, and more specifically, still, the embodiment depicted includes thirty-two part span inlet guide vanes **300**. Additionally, for the embodiment depicted, each of the plurality of part span inlet guide vanes **300** is spaced substantially evenly along a circumferential direction C of the turbofan engine **10**. More specifically, each of the plurality of part span inlet guide vanes **300** defines a circumferential spacing **318** with an adjacent part span inlet guide vane **300**, with the circumferential spacing **318** being substantially equal between each adjacent part span inlet guide vane **300**.
(143) Although not depicted, in certain exemplary embodiments, the number of part span inlet

guide vanes **300** may be equal to the number of fan blades **40** of the fan **38** of the turbofan engine **10**. In other embodiments, however, the number of part span inlet guide vanes **300** may be greater than the number of fan blades **40** (FIG. **15**) of the fan **38** (FIG. **15**) of the turbofan engine **10**, or alternatively, may be less than the number of fan blades **40** of the fan **38** of the turbofan engine **10**.

(144) Further, it should be appreciated, that in other exemplary embodiments, the turbofan engine **10** may include any other suitable number of part span inlet guide vanes **300** and/or circumferential spacing **318** of the part span inlet guide vanes **300**. For example, referring now briefly to FIG. **17**, an axial view of an inlet **60** to a turbofan engine **10** in accordance with another exemplary embodiment of the present disclosure is provided. For the embodiment of FIG. **17**, the turbofan engine **10** includes less than twenty part span inlet guide vanes **300**. More specifically, for the embodiment of FIG. **17**, the turbofan engine **10** includes at least eight part span inlet guide vanes **300**, or more specifically includes exactly eight part span inlet guide vanes **300**. Additionally, for the embodiment of FIG. **17** the plurality of part span inlet guide vanes **300** are not substantially evenly spaced along the circumferential direction C. For example, at least certain of the plurality of part span inlet guide vanes **300** defines a first circumferential spacing **318**A, while other of the plurality of part span inlet guide vanes **300** defines a second circumferential spacing **318**B. For the embodiment depicted, the first circumferential spacing **318**A is at least about twenty percent greater than the second circumferential spacing **318**B, such as at least about twenty-five percent greater such as at least about thirty percent greater, such as less than or equal to about two hundred percent greater. Notably, the circumferential spacing **318** refers to a mean circumferential spacing between adjacent part span inlet guide vanes **300**. The non-uniform circumferential spacing may, e.g., offset structure upstream of the part span inlet guide vanes **300**.

(145) Referring now back to FIG. **15**, each of the plurality of part span inlet guide vanes **300** is configured to pre-swirl an airflow **58** provided through the inlet **60** of the outer nacelle **50**, upstream of the plurality of fan blades **40** of the fan **38**. As discussed herein, pre-swirling the airflow **58** provided through the inlet **60** of the outer nacelle **50** prior to such airflow **58** reaching the plurality of fan blades **40** of the fan **38** may reduce separation losses and/or shock losses, allowing the fan **38** to operate with the relatively high fan tip speeds described above with less losses in efficiency.

(146) For example, referring first to FIG. **18**, a cross-sectional view of one part span inlet guide vane **300** along the span of the part span inlet guide vanes **300**, as indicated by Line **18-18** in FIG. **15**, is provided. As is depicted, the part span inlet guide vane **300** is configured generally as an airfoil having a pressure side **320** and an opposite suction side **322**, and extending between the leading edge **308** and the trailing edge **310** along a camber line **324**. Additionally, the part span inlet guide vane **300** defines a chord line **326** extending directly from the leading edge **308** to the trailing edge **310**. The chord line **326** of the part span inlet guide vane **300** defines an angle of attack **328** with respect to the longitudinal centerline **12** of the outer nacelle **50** (see, also, FIG. **15**). For example, the chord line **326** defines an angle of attack **328** with an airflow direction **329** of the airflow **58** through the inlet **60** of the nacelle **50**. Notably, for the embodiment depicted, the airflow direction **329** is substantially parallel to the axial direction A and the longitudinal centerline **12** of the outer nacelle **50** of the turbofan engine **10**. For the embodiment depicted, the angle of attack **328** at the location depicted along the IGV span **306** of the part span inlet guide vanes **300** is at least approximately five degrees and less than or equal to approximately thirty-five degrees. For example, in certain embodiments, the angle of attack **328** at the location depicted along the IGV span **306** of the part span inlet guide vane **300** may be between about ten degrees and about thirty degrees, such as between about fifteen degrees and about twenty-five degrees.

(147) Additionally, the part span inlet guide vane **300**, at the location depicted along the IGV span **306** of the part span inlet guide vane **300** defines a local swirl angle **330** at the trailing edge **310**. The “swirl angle” at the trailing edge **310** of the part span inlet guide vane **300**, as used herein, refers to an angle between the airflow direction **329** of the airflow **58** through the inlet **60** of the

outer nacelle **50** and a reference line **332** defined by a trailing edge section of the pressure side **320** of the part span inlet guide vane **300**. More specifically, the reference line **332** is defined by the aft twenty percent of the pressure side **320**, as measured along the chord line **326**. Notably, when the aft twenty percent the pressure side **320** defines a curve, the reference line **332** may be straight-line average fit of such curve (e.g., using least mean squares).

(148) Further, a maximum swirl angle **330** refers to the highest swirl angle **330** along the IGV span **306** of the part span inlet guide vane **300**. For the embodiment depicted, the maximum swirl angle **330** is defined proximate the radially outer end **302** of the part span inlet guide vane **300** (e.g., at the outer ten percent of the IGV span **306** of the part span inlet guide vanes **300**), as is represented by the cross-sectional view depicted in FIG. **18**. For the embodiment depicted, the maximum swirl angle **330** of each part span inlet guide vane **300** at the trailing edge **310** is between approximately five degrees and approximately thirty-five degrees. For example, in certain exemplary embodiments, the maximum swirl angle **330** of each part span inlet guide vane **300** at the trailing edge **310** may be between twelve degrees and twenty-five degrees.

(149) Moreover, for the embodiment of FIGS. **15**, **16**, and **18**, the local swirl angle **330** increases from the radially inner end **304** to the radially outer end **302** of each part span inlet guide vane **300**. For example, referring now also to FIG. **19**, a cross-sectional view of a part span inlet guide vane **300** at a location radially inward from the cross-sectional view in FIG. **18**, as indicated by Line **19-19** in FIG. **15**, is provided. As is depicted in FIG. **19**, and as stated above, the part span inlet guide vane **300** defines the pressure side **320**, the suction side **322**, the leading edge **308**, the trailing edge **310**, the camber line **324**, and the chord line **326**. Further, the angle of attack **328** defined by the chord line **326** and the airflow direction **329** of the airflow **58** through the inlet **60** of the outer nacelle **50** at the location along the IGV span **306** depicted in FIG. **19** is less than the angle of attack **328** at the location along the IGV span **306** depicted in FIG. **18** (e.g., may be at least about twenty percent less, such as at least about fifty percent less, such as less than or equal to about one hundred percent less). Additionally, the part span inlet guide vane **300** defines a local swirl angle **330** at the trailing edge **310** at the location along the IGV span **306** of the part span inlet guide vane **300** proximate the inner end **304**, as depicted in FIG. **19**. As stated above, the local swirl angle **330** increases from the radially inner end **304** to the radially outer end **302** of each part span inlet guide vanes **300**. Accordingly, the local swirl angle **330** proximate the outer end **302** (see FIG. **18**) is greater than the local swirl angle **330** proximate the radially inner end **304** (see FIG. **19**; e.g., the radially inner ten percent of the IGV span **306**). For example, the local swirl angle **330** may approach zero degrees (e.g., may be less than about five degrees, such as less than about two degrees) at the radially inner end **304**.

(150) Notably, including part span inlet guide vanes **300** of such a configuration may reduce an amount of turbulence at the radially inner end **304** of each respective part span inlet guide vane **300**. Additionally, such a configuration may provide a desired amount of pre-swirl at the radially outer ends of the plurality of fan blades **40** of the fan **38** (where the speed of the fan blades **40** is the greatest) to provide a desired reduction in flow separation and/or shock losses that may otherwise occur due to a relatively high speed of the plurality of fan blades **40** at the fan tip **72***s* during operation of the turbofan engine **10**.

(151) Referring now to FIG. **20**, a close-up, schematic view of a fan section **14** and forward end of a turbomachine **16** of a turbofan engine **10** in accordance with another exemplary embodiment of the present disclosure is provided. The exemplary turbofan engine **10** of FIG. **20** may be configured in a similar manner as the exemplary turbofan engine **10** of FIGS. **14** through **16** and **18** through **19**.

(152) For the exemplary embodiment of FIG. **20**, the turbofan engine **10** further includes a variable pitch mechanism **340** that is in communication with the inlet pre-swirl feature, e.g., configured as the plurality of part span inlet guide vanes **300**. The variable pitch mechanism **340** is configured to transition the inlet pre-swirl feature, e.g., configured as the plurality of part span inlet guide vanes

300, between a first angle with respect to a longitudinal centerline **12** of turbofan engine **10** and a second angle with respect to the longitudinal centerline **12** of the turbofan engine **10**. In such a manner, it will be appreciated that each of the plurality of part span inlet guide vanes **300** defines a pitch axis P about which it is configured to rotate. It is contemplated that the variable pitch mechanism **340** may include, for example, a stepper motor, a torque motor, or similar drive component.

(153) In such a manner, each of the part span inlet guide vanes **300** of the present disclosure is transitionable between the first angle with respect to the longitudinal centerline **12** of the outer nacelle **50** and the second angle, e.g., an angle of attack **328** or a local swirl angle **330**, with respect to the longitudinal centerline **12** of the outer nacelle **50**, wherein the first angle and the second angle are different. The first angle may be similar to the angle depicted in the embodiment of FIG. **18** (e.g., the angle of attack **328** or the local swirl angle **330**), and the second angle may be similar to the angle depicted in the embodiment of FIG. **19** (e.g., the angle of attack **328** or the local swirl angle **330**; when the same part span inlet guide vane **300** is viewed at the same location along its span).

(154) In other words, an angle of the part span inlet guide vanes **300** of the present disclosure can be varied during operation of the turbofan engine **10**. In certain exemplary embodiments it is contemplated that the second angle is at least about 2% greater/less than the first angle. In other exemplary embodiments, it is contemplated that the second angle is at least about 5% greater/less than the first angle. In other exemplary embodiments, it is contemplated that the second angle is at least about 30% greater/less than the first angle.

(155) In the present disclosure, the angle of the part span inlet guide vanes **300** may be variable in order to, e.g., match the swirl imparted to the incoming air to the airspeed of the aircraft and the rotational speed of the fan **38** such that the angular velocity of the air as it approaches the fan blade **40** corresponds more closely with an angular velocity of the fan blade **40**. This may reduce a potential of the fan **38** to surge/stall. The faster the fan **38** rotates, the more swirl that needs to be imparted by the part span inlet guide vanes **300**. As the airspeed of the aircraft increases, the time that it takes for the incoming air to pass from the part span inlet guide vanes **300** to the leading edge **314** of the fan **38** decreases, and as such the necessary amount of swirl decreases proportionately. As such, a maximum imparted swirl may be required when the turbofan engine **10** is at a maximum thrust with a stationary aircraft, just prior to beginning a takeoff operation.

(156) For example, each of the part span inlet guide vanes **300** of the present disclosure could transition from a higher angle (FIG. **18**) to a lower angle (FIG. **19**; when viewed at a common location along a span of the part span inlet guide vane **300**) during operation of the turbofan engine **10**. Furthermore, each of the part span inlet guide vanes **300** of the present disclosure could transition from a lower angle (FIG. **18**) to a higher angle (FIG. **19**; when viewed at a common location along a span of the part span inlet guide vane **300**) during operation of the turbofan engine **10**.

(157) Referring still to FIG. **20**, in exemplary embodiments, the plurality of part span inlet guide vanes **300** may further be configured to provide a compensation airflow **360** to a trailing edge **310** of the plurality of part span inlet guide vanes **300** to minimize a wake of the part span inlet guide vanes **300** and/or to address crosswind concerns for the turbofan engine **10**.

(158) For example, as in other embodiments of the present disclosure, in order to realize a benefit achievable through inclusion of the plurality of outlet guide vanes **52** configured in the forward swept configuration, the turbofan engine **10** further includes a relatively short inlet. In particular, the outer nacelle **50** defines an inlet length L and the fan **38** defines a fan diameter D (equal to twice a fan radius **45**). The ratio of the inlet length L to the fan diameter D is less than or equal to 0.5 in the embodiment shown. In such a manner, the inlet of the outer nacelle **50** may be less capable of straightening out a crosswind experienced by the turbofan engine **10** during certain operating conditions (e.g., taxi and takeoff). In certain exemplary embodiments, the plurality of

inlet guide vanes **300** may assist with addressing crosswind concerns of the turbofan engine **10** by, e.g., varying a pitch of one or more of the plurality of part span inlet guide vanes **300**, introducing compensation airflow **360** through one or more of the plurality of part span inlet guide vanes **300**, or both.

(159) In particular, the exemplary turbofan engine **10** of FIG. **20** includes a compensation air supply assembly **362**, which generally includes a compensation air supply duct **364** defining an inlet **366** in airflow communication with the high pressure air source. For the embodiment depicted, the high pressure air source is a compressor section of the turbomachine **16** of the turbofan engine **10**. For example, the compensation air supply duct **364** may be configured to receive bleed air from the compressor section.

(160) Notably, however, in other embodiments, the compensation air supply duct **364** may instead receive high pressure air from any other suitable high pressure air source. For example, in other exemplary embodiments, the high pressure air source may instead be the bypass airflow duct **56** at a location downstream of the plurality of fan blades **40** of the fan **38**. Additionally, in one or more these embodiments, the compensation air supply assembly **362** may further include an air compressor **367** (depicted in phantom) configured to increase a pressure of the compensation airflow **360** through the compensation air supply duct **364**.

(161) Further, although the compensation air supply duct **364** is depicted as a single, continuous, and separate compensation air supply duct **364**, in other embodiments, the composition air supply duct **364** may have any other suitable configuration. For example, the compensation air supply duct **364** may be formed of a plurality of sequential ducts, may be formed integrally with other components of the turbofan engine **10**, and/or may be split off into a plurality of parallel airflow ducts to provide compensation airflow **360** to each of the plurality of part span inlet guide vanes **300**.

(162) Further, the compensation air supply duct **364** extends through at least one of the plurality of part span inlet guide vanes **300**, and provides a cavity **368** of the part span inlet guide vane **300** with the high pressure composition airflow **360**. As is depicted, each of the plurality of part span inlet guide vanes **300** for the embodiment depicted further defines a trailing edge opening **370**, which is in airflow communication with the cavity **368**, and thus is in airflow communication with the compensation air supply duct **364** of the compensation air supply assembly **362**. Accordingly, with such a configuration, the high pressure composition airflow **360** may be provided from the compensation air supply assembly **362** to the cavity **368** of the part span inlet guide vane **300**, and further through the trailing edge opening **370** of the part span inlet guide vane **300** during operation of the turbofan engine **10** to, e.g., reduce a wake formed by the respective part span inlet guide vane **300**, address crosswind issues experienced by the turbofan engine **10**, assist with pre-swirling activities, etc.

(163) Although described as a “cavity”, in other embodiments the cavity **368** may be configured as any suitable opening or passage within the part span inlet guide vane **300** to allow a flow of air therethrough. Additionally, in other exemplary embodiments, the plurality of part span inlet guide vanes **300** may instead include any other suitable manner of pneumatically reducing the wake of the respective part span inlet guide vanes **300**. For example, in other exemplary embodiments, the trailing edge opening **370** of each part span inlet guide vane **300** may instead be configured as, e.g., a plurality of trailing edge of openings **370** spaced, e.g., along a span **306** of the respective part span inlet guide vane **300** at the trailing edge **310**.

(164) Referring still to FIG. **20** and now also to FIG. **21**, FIG. **21** provides an axial view of the inlet **60** to the turbofan engine **10** of FIG. **20**. In the exemplary embodiment depicted, the outer nacelle **50** includes a top portion **410**, a bottom portion **412**, a first side portion **414**, and a second side portion **416**. In such an embodiment, the plurality of part span inlet guide vanes **300** in the top portion **410** are in sector A, the plurality of part span inlet guide vanes **300** in the first side portion **414** are in sector B, the plurality of part span inlet guide vanes **300** in the second side portion **416**

are in sector C, and the plurality of part span inlet guide vanes **300** in the bottom portion **412** are in sector D. Furthermore, in such an exemplary embodiment, the transitioning of an angle of the part span inlet guide vanes **300**, with respect to the longitudinal centerline **12** in each sector A, B, C, and D, is separately controlled.

(165) In particular, for the exemplary embodiment depicted, the turbofan engine **10** further includes a first pitch change mechanism **340**A operably coupled to the plurality of part span inlet guide vanes **300** in sector A, a second pitch change mechanism **340**B operably coupled to the plurality of part span inlet guide vanes **300** in sector B, a third pitch change mechanism **340**C operably coupled to the plurality of part span inlet guide vanes **300** in section C, and a fourth pitch change mechanism **340**D operably coupled to the plurality of part span inlet guide vanes **300** in sector D.

(166) Each of the plurality of pitch change mechanisms **340**A, **340**B, **340**C, **340**D is in operable communication with a controller **500** of the turbofan engine **10**. In such a manner, the controller **500** is configured to control one or more of the plurality of pitch change mechanisms **340**A, **340**B, **340**C, **340**D independently, such that the plurality of part span the guide vanes **300** in at least one of sectors A, B, C, or D may be independently controlled relative to the plurality of part span guide vanes **300** in at least one of sectors A, B, C, or D. For example, the plurality of part span inlet guide vanes **300** may generally include a first part span inlet guide vane **300** (e.g., a first pre-swirl feature) located in sector B transitional between a first angle and a second angle with respect to the longitudinal centerline **12** (see, e.g., FIGS. **18** and **19**, when viewed at a common location along a span of the first part span inlet guide vane **300**). The plurality of part span inlet guide vanes **300** may further include a second part span inlet guide vane **300** (e.g., a second pre-swirl feature) located in sector C transitional between a third angle and a fourth angle with respect to the longitudinal centerline **12** of the turbofan engine **10** (see, e.g., FIGS. **18** and **19**, when viewed at a common location along a span of the second part span inlet guide vane **300**). The first part span inlet guide vane **300** and second part span inlet guide vane **300** may be independently controlled by the controller **500** (e.g., through the pitch change mechanisms **340**B, **340**C).

(167) For example, an aircraft engine inlet **60** is exposed to the atmosphere, and to wind conditions that may be from any direction. Furthermore, since the turbofan engine **10** is typically shielded on the side that is closer to a fuselage of the aircraft, the wind conditions may be stronger on a side that is away from the fuselage of the aircraft. The aircraft may also be subject to up/downdrafts. As the wind can generally be expected to blow in a singular direction across the face of the engine **10**, each of the part span inlet guide vanes **300** may be at a different angle with respect to the wind. As such the effect of a cross wind on the incoming air is a function of the angle of the part span inlet guide vane **300** with respect to the cross wind. It is desirable to impart a swirl to the incoming air that accounts for the cross wind velocity at the particular clock position of the individual part span inlet guide vanes **300**. As such individualized control of the angle of the part span inlet guide vanes **300** is desirable.

(168) Notably, the issue of crosswind may be enhanced by virtue of the short inlet (L/D less than 0.5) of the turbofan engine **10** depicted.

(169) As the complexity of the part span inlet guide vane control system increases, the weight of the system increases, which has an effect on the total engine/aircraft efficiency. Therefore, it may be desirable to group the part span inlet guide vanes **300** into clusters to minimize the weight of the total system, while still being able to control the angle of the part span inlet guide vanes **300** of differing sectors of the inlet, such as left and right sectors B and C, respectively, or top and bottom sectors A and D, respectively.

(170) Referring still to FIGS. **20** and **21**, it will be appreciated that the turbofan engine **10** further includes one or more sensors **502** configured to sense data indicative of a condition of the turbofan engine **10**, such as a condition of the plurality of fan blades **40**. The condition of the turbofan engine **10** may be a forward speed of the turbofan engine **10**, a crosswind experience by the inlet **60** to the turbofan engine **10**, etc. The condition of the plurality of fan blades **40** may be a rotational

speed of the plurality of fan blades **40**, a position of the plurality of fan blades **40** (along the circumferential direction C), etc. In particular, for the embodiment of FIG. **20**, the turbofan engine **10** includes a first sensor **502**A, which may be a crosswind sensor configured to sense data indicative of a crosswind experience by the turbofan engine **10**, such as by the inlet **60** of the turbofan engine **10**, and a second sensor **502**B configured as a blade passing sensor configured to sense data indicative of a rotational speed and/or of a position of the plurality of fan blades **40**. The first sensor **502**A and the second sensor **502**B are operably coupled to the controller **500** for providing data to the controller **500**.

(171) In certain exemplary aspects, the controller **500** may be configured to receive input data indicating the condition of the plurality of fan blades **40**, a condition of the turbofan engine **10**, or both, and in response to the data control one or more aspects of the plurality of part span inlet guide vanes **300**.

(172) For example, in one exemplary embodiment, in response to the received data, the controller **500** may be configured to control a pitch of the plurality of part span the guide vanes **300**. Controlling the pitch of the plurality of part span inlet guide vanes **300** may include varying a pitch of a first part span inlet guide vane **300** relative to a second part span inlet guide vane **300**. For example, such an exemplary aspect may include controlling one of the plurality of pitch change mechanisms **340**A, **340**B, **340**C, **340**D relative to one or more of the remaining pitch change mechanisms **340**A, **340**B, **340**C, **340**D (see FIG. **21**).

(173) Further for example, in another exemplary embodiment, in response to the received data, the controller **500** may be configured to modulate a high-pressure airflow through the openings at the trailing edges **370** of the plurality of part span inlet guide vanes **300**. Briefly, as is depicted schematically in FIG. **20**, the turbofan engine **10** may include one or more valves **504** operably coupled to the controller **500** allowing for such a modulation. Moreover, referring now briefly to FIG. **21**, the turbofan engine **10** may have a plurality of valves **504** operably coupled to the controller **500** allowing for such a modulation. In particular, the turbofan engine **10** depicted schematically includes a first valve **504**A, a second valve **504**B, a third valve **504**C, and a fourth valve **504**D.

(174) Notably, in certain exemplary aspects, modulating the high-pressure airflow through the openings **370** at the trailing edges **310** of the plurality of part span inlet guide vanes **300** may include modulating the high-pressure airflow through a first opening **370** at a first trailing edge **310** of a first part span the guide vane **300** relative to a high-pressure airflow through a second opening **370** at a second trailing edge **310** of a second part span of the guide vane **300**. For example, as is depicted schematically in FIG. **21**, the turbofan engine **10** may include a first valve **504**A fluidly connecting the high pressure air source to the plurality of part span inlet guide vanes **300** in sector A, a second valve **504**B fluidly connecting the high pressure air source to the plurality of part span inlet guide vanes **300** in sector B, a third valve **504**C fluidly connecting the high pressure air source to the plurality of part span inlet guide vanes **300** in sector C, and a fourth valve **504**D fluidly connecting the high pressure air source to the plurality of part span inlet guide vanes **300** in sector D. With such exemplary aspect, modulating the airflow may include controlling one or more of the valves **504**A, **504**B, **504**C, **504**D relative to one or more of the remaining valves **504**A, **504**B, **504**C, **504**D (see FIG. **21**).

(175) As noted, the exemplary controller **500** depicted in FIG. **22** is configured to receive the data sensed from various data sources, such as one or more sensors **502** (FIG. **20**) (e.g., sensors **502**A, **502**B for the embodiment shown) and, e.g., may make control decisions for the turbofan engine **10** (FIG. **20**) based on the received data.

(176) In one or more exemplary embodiments, the controller **500** depicted in FIG. **22** may be a stand-alone controller **500** for the turbofan engine **10**, or alternatively, may be integrated into one or more other controllers for the turbofan engine **10**, a controller for an aircraft including the turbofan engine **10**, etc.

(177) Referring particularly to the operation of the controller **500**, in at least certain embodiments, the controller **500** can include one or more computing device(s) **510**. The computing device(s) **510** can include one or more processor(s) **510**A and one or more memory device(s) **510**B. The one or more processor(s) **510**A can include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, and/or other suitable processing device. The one or more memory device(s) **510**B can include one or more computer-readable media, including, but not limited to, non-transitory computer-readable media, RAM, ROM, hard drives, flash drives, and/or other memory devices.
(178) The one or more memory device(s) **510**B can store information accessible by the one or more processor(s) **510**A, including computer-readable instructions **510**C that can be executed by the one or more processor(s) **510**A. The instructions **510**C can be any set of instructions that when executed by the one or more processor(s) **510**A, cause the one or more processor(s) **510**A to perform operations. In some embodiments, the instructions **510**C can be executed by the one or more processor(s) **510**A to cause the one or more processor(s) **510**A to perform operations, such as any of the operations and functions for which the controller **500** and/or the computing device(s) **510** are configured, the operations for turbofan engine **10**, as described herein, and/or any other operations or functions of the one or more computing device(s) **510**. The instructions **510**C can be software written in any suitable programming language or can be implemented in hardware. Additionally, and/or alternatively, the instructions **510**C can be executed in logically and/or virtually separate threads on the one or more processor(s) **510**A. The one or more memory device(s) **510**B can further store data **510**D that can be accessed by the one or more processor(s) **510**A. For example, the data **510**D can include data indicative of power flows, data indicative of engine/aircraft operating conditions, and/or any other data and/or information described herein.
(179) The computing device(s) **510** can also include a network interface **510**E used to communicate, for example, with the other components of the turbofan engine **10**. For example, in the embodiment depicted, as noted above, the turbofan engine includes one or more sensors **502** for sensing data indicative of one or more parameters of the turbofan engine **10**. The controller **500** is operably coupled to the one or more sensors **502** through, e.g., the network interface, such that the controller **500** may receive data indicative of various operating parameters sensed by the one or more sensors **502** during operation. Further, for the embodiment shown the controller **500** is operably coupled to, e.g., the pitch change mechanism(s) **340**, the valve(s) **504**, etc. In such a manner, the controller **500** may be configured to control the pitch change mechanism(s) **340**, the valve(s) **504**, etc. in response to, e.g., the data sensed by the one or more sensors **502**.
(180) The network interface **510**E can include any suitable components for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controllers, antennas, and/or other suitable components.
(181) The technology discussed herein makes reference to computer-based systems and actions taken by and information sent to and from computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. For instance, processes discussed herein can be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications can be implemented on a single system or distributed across multiple systems. Distributed components can operate sequentially or in parallel.
(182) Referring now to an additional and/or alternative exemplary configuration of the present disclosure, it will be appreciated that the demand for more aerodynamically efficient and compact turbofan engines remains strong. Turbofan engines have traditionally sought to relatively evenly distribute compression loads between a booster (also referred to as an LP compressor) and a high pressure compressor (HP compressor), aiming for operational balance.
(183) However, the inventors of the present disclosure have found that pursuing this balance may

lead to certain constraints. For example, previous configurations have relied on a reduction gearbox to moderate fan speeds while allowing a drive turbine, such as a low pressure turbine (LP turbine), and the booster to rotate more quickly. Such a configuration allows for the booster to contribute more significantly to an overall compression of a compressor section of the turbofan engine. While effective, this approach might result in underutilized turbine efficiency.
(184) In a departure from this norm, the inventors of the present disclosure have found an advantage associated with leveraging a high pressure ratio (PR) core/HP compressor, in combination with a reduction gearbox positioned between the LP turbine and the booster such that the booster may rotate more slowly than the LP turbine. This strategic arrangement may allow for the HP compressor to provide a more substantial portion of the overall compression of the compressor section of the turbofan engine. Consequently, more work may then be extracted by a high pressure turbine (HP turbine), thereby reducing a burden on the LP turbine. The inclusion of a geared connection between the LP turbine and the booster/LP compressor also allows for the LP turbine to rotate more quickly, potentially allowing for a reduction in rotor stage count and/or an expansion in the LP turbine's radius to reduce an overall length of the turbomachine of the turbofan engine.
(185) Further, incorporating forward-swept OGVs to this arrangement may further allow for a reduction in a length of an outer nacelle of the turbofan engine. Thus, by integrating the forward-swept OGVs with the above configuration, the inventors have found that an overall length of the turbofan engine may be reduced, while also improving aerodynamic performance and engine efficiency.
(186) Referring now to FIG. **23**, a turbofan engine **10** in accordance with another exemplary embodiment of the present disclosure is provided. The exemplary turbofan engine **10** of FIG. **23** may be configured in a similar manner as one or more of the exemplary turbofan engines **10** described above, and the same or similar numbering refers to the same or similar parts.
(187) For example, the exemplary turbofan engine **10** of FIG. **23** generally includes a fan section **14** having a fan **38** with a plurality of fan blades **40**, a turbomachine **16** drivingly coupled to the fan **38** and having an outer casing **18**, and an outer nacelle **50** surrounding the fan **38** and at least a portion of the turbomachine **16**. Further, the exemplary turbofan engine **10** includes an outlet guide vane **52** extending between the turbomachine **16** and the outer nacelle **50** at a location downstream of the plurality of fan blades **40**. The outlet guide vane **52** defines a base **70** and a tip **72** and is forward swept from the base **70** to the tip **72**. More specifically, the turbofan engine **10** includes a plurality of outlet guide vanes **52** configured in such a manner.
(188) Further, the exemplary turbofan engine **10** of FIG. **23** includes a shortened inlet. More specifically, the outer nacelle **50** defines an inlet length L and the fan **38** defines a fan diameter D (equal to twice a fan radius **45**). A ratio of the inlet length L to the fan diameter D is less than or equal to 0.5.
(189) In such a manner, will be appreciated that the outer nacelle **50** may define a relatively short axial length, particularly in view of the outlet guide vanes **52** being forward swept from the respective base(es) **70** to the respective tip(s) **72**.
(190) Further, the turbomachine **16** of FIG. **23** is arranged in a manner to also allow for a shortened overall length of the turbomachine **16** for the size of the turbofan engine **10**. In particular, the exemplary turbomachine **16** includes a compressor section having an LP compressor **22** and an HP compressor **24**, a combustion section **26**, and a turbine section having an HP turbine **28** and an LP turbine **30**.
(191) The turbomachine **16** further includes a power gearbox **46** (also referred to as a reduction gearbox). For the embodiment depicted, the LP turbine **30** is drivingly coupled to the LP compressor **22** across the power gearbox **46**. More specifically, the turbomachine **16** includes an LP shaft **36** rotatable with the LP turbine **30** and a fan shaft **110** rotatable with the fan **38**. The fan shaft **110** is driven by the LP shaft **36** across the power gearbox **46**, and the LP compressor **22** is driven

by the fan **38**. In such manner, it will be appreciated that the LP compressor **22** is directly rotatable with the fan **38** (e.g., the LP compressor **22** rotates at the same rotational speed as the fan **38**).
(192) Such a configuration allows for a beneficial packaging of the power gearbox **46**. In particular, for the embodiment depicted, the power gearbox **46** is aligned with the LP compressor **22** along the longitudinal centerline **12**, and more specifically is aligned with a downstream-most stage of LP compressor rotor blades of the LP compressor **22** along the longitudinal centerline **12**. Moreover, it will be appreciated that such a configuration allows for the LP compressor **22** to be moved forward, and accordingly for an inter-compressor frame **108** of the turbomachine **16** to be moved forward. Accordingly, it will further be appreciated from the view of FIG. **23** that the base **70** of the outlet guide vane **52** is coupled to the inter-compressor frame **108** and is aligned with the inter-compressor frame **108** along the longitudinal centerline **12**. Such configuration may provide for more direct support of the outlet guide vane **52**.
(193) Further, with such configuration, an aft side of the power gearbox **46** overlaps with the inter-compressor frame **108** of the turbomachine **16**. Such a configuration may provide for a more compact compressor section.
(194) Further, it will be appreciated that with such configuration the LP compressor **22** may rotate at a relatively low rotational speed, and thus may define a relatively low LP compressor pressure ratio PRLPC. Accordingly, for the embodiment depicted, the HP compressor **24** is configured to define a relatively high HP compressor pressure ratio PRHPC. For example, in certain exemplary aspects, the HP compressor pressure ratio PRHPC defined by the HP compressor **24** may be at least 22 and less than or equal to 30 during an operating condition of the turbofan engine **10**. The HP compressor pressure ratio PRHPC may refer to a pressure ratio of an airflow exiting the HP compressor **24** to a pressure of an airflow entering the HP compressor **24** during the operating condition of the turbofan engine **10**.
(195) Notably, with the LP compressor **22** being driven by the LP shaft **36** across the power gearbox **46**, the LP turbine **30** can rotate at a relatively high speed, allowing for a shortening of the LP turbine **30**. In particular, for the embodiment depicted, the LP turbine **30** is configured as a three-stage LP turbine **30** (having three stages of LP turbine rotor blades **31**). Such may result in a relatively short LP turbine **30**, further allowing for a reduction in overall length of the turbofan engine **10**.
(196) Further aspects are provided by the subject matter of the following clauses:
(197) A turbofan engine defining an axial direction and a longitudinal centerline along the axial direction, the turbofan engine comprising: a fan section having a fan; a turbomachine drivingly coupled to the fan, the turbomachine comprising an outer casing; an outer nacelle surrounding the fan and at least a portion of the turbomachine, the turbofan engine defining a bypass passage between the outer nacelle and the turbomachine; an outlet guide vane extending between the turbomachine and the outer nacelle, the outlet guide vane defining a base and a tip and being forward swept from the base to the tip; and an acoustic treatment attached to or integrated with the outer casing at a location aligned with the outlet guide vane along the longitudinal centerline.
(198) The turbofan engine of the preceding clause, wherein the outlet guide vane defines a junction with the outer casing at a leading edge of the outlet guide vane, and wherein the acoustic treatment is located at the junction, forward of the junction, or both.
(199) The turbofan engine of any preceding clause, wherein the turbomachine defines an inlet, and wherein the acoustic treatment extends along the axial direction at least 50% of a distance along the axial direction from the inlet to the junction.
(200) The turbofan engine of any preceding clause, wherein the acoustic treatment comprises a perforated sheet and a hollow body.
(201) The turbofan engine of any preceding clause, wherein the outlet guide vane comprises an acoustic treatment on a pressure side, on a suction side, or both.
(202) The turbofan engine of any preceding clause, wherein the outlet guide vane defines an OGV

reference line extending from an inner junction between the outlet guide vane and the turbomachine at a leading edge of the outlet guide vane and an outer junction between the outlet guide vane and the outer nacelle at the leading edge of the outlet guide vane, wherein the turbofan engine defines a radial reference line extending perpendicularly from the longitudinal centerline, and wherein an angle between the OGV reference line and the radial reference line is at least 5 degrees and less than or equal to 45 degrees.
(203) The turbofan engine of any preceding clause, wherein the angle between the OGV reference line and the radial reference line is at least 15 degrees and less than or equal to 35 degrees.
(204) The turbofan engine of any preceding clause, wherein the outer nacelle comprises an outer attachment groove located at a trailing edge of the outlet guide vane.
(205) The turbofan engine of any preceding clause, wherein the turbomachine comprises an inner attachment groove located at the trailing edge of the outlet guide vane, and wherein the inner attachment groove is located aft of the outer attachment groove.
(206) The turbofan engine of any preceding clause, wherein the outer nacelle defines an inlet length, wherein the fan defines a fan diameter, and wherein a ratio of the inlet length to the fan diameter is less than or equal to 0.5.
(207) The turbofan engine of any preceding clause, wherein the turbomachine comprises a reduction gearbox and a compressor section having a low pressure compressor, wherein the outlet guide vane defines an inner junction between the outlet guide vane and the turbomachine at the leading edge of the outlet guide vane, and wherein the inner junction is aligned with the low pressure compressor along the longitudinal centerline.
(208) The turbofan engine of any preceding clause, wherein the compressor section further comprises a high pressure compressor, wherein the turbomachine comprises a compressor forward frame forward of the low pressure compressor and an inter-compressor frame between the high pressure compressor and the low pressure compressor, wherein the turbomachine further comprises a shell frame located between the compressor forward frame and the inter-compressor frame, and wherein the outlet guide vane is coupled to the shell frame.
(209) The turbofan engine of any preceding clause, wherein the turbomachine comprises an A-sump cone and a load reduction device integrated into the sump cone or into an attachment of the sump cone.
(210) The turbofan engine of any preceding clause, wherein the outlet guide vane is a first outlet guide vane of a plurality of outlet guide vanes, wherein each outlet guide vane of the plurality of outlet guide vanes defines a base and a tip and is forward swept from the base to the tip.
(211) The turbofan engine of any preceding clause, wherein the acoustic treatment extends along a circumferential direction of the turbofan engine.
(212) The turbofan engine of any preceding clause, wherein the acoustic treatment extends substantially completely along a circumferential direction of the turbofan engine.
(213) The turbofan engine of any preceding clause, wherein the turbofan engine defines a bypass ratio equal to a ratio of a mass flowrate of airflow through the bypass passage to a mass flowrate of an airflow through an inlet of the turbomachine during cruise operations, wherein the bypass ratio is at least 5:1 and less than or equal to 20:1.
(214) The turbofan engine of any preceding clause, wherein the fan defines a fan diameter greater than or equal to four feet and less than or equal to 18 feet.
(215) The turbofan engine of any preceding clause, wherein the fan is a single stage fan.
(216) The turbofan engine of any preceding clause, wherein the turbofan defines a maximum rated thrust at sea level greater than or equal to 30,000 pounds and less than or equal to 120,000 pounds.
(217) A turbofan engine defining an axial direction and a longitudinal centerline along the axial direction, the turbofan engine comprising: a fan section having a fan; a turbomachine drivingly coupled to the fan, the turbomachine comprising an outer casing; an outer nacelle surrounding the fan and at least a portion of the turbomachine; an outlet guide vane extending between the

turbomachine and the outer nacelle, the outlet guide vane defining a base and a tip and being forward swept from the base to the tip; and an accessory gearbox positioned at least partially inward of the outer casing of the turbomachine.
(218) The turbofan engine of any preceding clause, wherein the accessory gearbox is positioned entirely within the outer casing of the turbomachine.
(219) The turbofan engine of any preceding clause, wherein the accessory gearbox includes a first portion and a second portion, wherein the first portion is positioned within the outer casing of the turbomachine, and wherein the second portion is positioned outside of the outer casing.
(220) The turbofan engine of any preceding clause, wherein the accessory gearbox comprises a drive assembly extending from the first portion to the second portion.
(221) The turbofan engine of any preceding clause, further comprising: a strut extending between the turbomachine and the outer nacelle, and wherein the drive assembly extends into the strut.
(222) The turbofan engine of any preceding clause, wherein the second portion is positioned in the outer nacelle.
(223) The turbofan engine of any preceding clause, wherein the second portion is positioned in the strut.
(224) The turbofan engine of any preceding clause, wherein the drive assembly comprises a drive shaft, a hydraulic drive, or a combination thereof.
(225) The turbofan engine of any preceding clause, wherein the outlet guide vane defines an OGV reference line extending from an inner junction between the outlet guide vane and the turbomachine at a leading edge of the outlet guide vane and an outer junction between the outlet guide vane and the outer nacelle at the leading edge of the outlet guide vane, wherein the turbofan engine defines a radial reference line extending perpendicularly from the longitudinal centerline, and wherein an angle between the OGV reference line and the radial reference line is at least 5 degrees and less than or equal to 45 degrees.
(226) The turbofan engine of any preceding clause, wherein the angle between the OGV reference line and the radial reference line is at least 15 degrees and less than or equal to 35 degrees.
(227) The turbofan engine of any preceding clause, wherein the outer nacelle comprises an outer attachment groove located at a trailing edge of the outlet guide vane.
(228) The turbofan engine of any preceding clause, wherein the turbomachine comprises an inner attachment groove located at the trailing edge of the outlet guide vane, and wherein the inner attachment groove is located aft of the outer attachment groove.
(229) The turbofan engine of any preceding clause, wherein the outer nacelle defines an inlet length, wherein the fan defines a fan diameter, and wherein a ratio of the inlet length to the fan diameter is less than or equal to 0.5.
(230) The turbofan engine of any preceding clause, wherein the outlet guide vane is a first outlet guide vane of a plurality of outlet guide vanes, wherein each outlet guide vane of the plurality of outlet guide vanes defines a base and a tip and is forward swept from the base to the tip.
(231) The turbofan engine of any preceding clause, wherein the turbofan engine defines a bypass ratio equal to a ratio of a mass flowrate of airflow through the bypass passage to a mass flowrate of an airflow through an inlet of the turbomachine during cruise operations, wherein the bypass ratio is at least 5:1 and less than or equal to 20:1.
(232) The turbofan engine of any preceding clause, wherein the fan defines a fan diameter greater than or equal to four feet and less than or equal to 18 feet.
(233) The turbofan engine of any preceding clause, wherein the fan is a single stage fan.
(234) The turbofan engine of any preceding clause, wherein the turbofan defines a maximum rated thrust at sea level greater than or equal to 30,000 pounds and less than or equal to 120,000 pounds.
(235) The turbofan engine of any preceding clause, further comprising: an acoustic treatment attached to or integrated with the outer casing at a location aligned with the outlet guide vane along the longitudinal centerline.

(236) The turbofan engine of any preceding clause, wherein the outlet guide vane defines a junction with the outer casing at a leading edge of the outlet guide vane, and wherein the acoustic treatment is located at the junction, forward of the junction, or both.
(237) A turbofan engine defining an axial direction and a longitudinal centerline along the axial direction, the turbofan engine comprising: a fan section having a fan, the fan comprising a plurality of fan blades; a turbomachine drivingly coupled to the fan, the turbomachine comprising an outer casing; an outer nacelle surrounding the fan and at least a portion of the turbomachine; an outlet guide vane extending between the turbomachine and the outer nacelle at a location downstream of the plurality of fan blades, the outlet guide vane defining a base and a tip and being forward swept from the base to the tip; and an inlet pre-swirl feature attached to or integrated into the outer nacelle at a location upstream of the plurality of fan blades.
(238) The turbofan engine of any preceding clause, wherein the inlet pre-swirl feature is transitionable between a first angle with respect to the longitudinal centerline and a second angle with respect to the longitudinal centerline.
(239) The turbofan engine of any preceding clause, wherein the inlet pre-swirl feature is a first inlet pre-swirl feature transitionable between a first angle with respect to the longitudinal centerline and a second angle with respect to the longitudinal centerline, wherein the turbofan engine further comprises: a second inlet pre-swirl feature located upstream of the plurality of fan blades, the second inlet pre-swirl feature attached to or integrated into the outer nacelle, wherein the second inlet pre-swirl feature is transitionable between a third angle with respect to the longitudinal centerline of the turbofan engine and a fourth angle with respect to the longitudinal centerline of the turbofan engine; wherein the first inlet pre-swirl feature and the second inlet pre-swirl feature are independently controlled.
(240) The turbofan engine of any preceding clause, wherein the inlet pre-swirl feature includes a leading edge, a trailing edge, and defines an opening at the trailing edge.
(241) The turbofan engine of any preceding clause, further comprising: a controller having one or more processors and one or more memory devices, the one or more memory devices storing instructions that when executed by the one or more processors cause the one or more processors to perform operations, in performing the operations, the one or more processors are configured to: receive an input indicating a condition of the plurality of fan blades; and in response to the condition of the plurality of fan blades, modulate a high pressure airflow through the opening at the trailing edge of the inlet pre-swirl feature during operation of the turbofan engine.
(242) The turbofan engine of any preceding clause, wherein the inlet pre-swirl feature comprises a part-span inlet guide vane.
(243) The turbofan engine of any preceding clause, wherein the part span inlet guide comprises an acoustic treatment.
(244) The turbofan engine of any preceding clause, wherein the outer nacelle defines an inlet length, wherein the fan defines a fan diameter, and wherein a ratio of the inlet length to the fan diameter is less than or equal to 0.5.
(245) The turbofan engine of any preceding clause, wherein The turbofan engine of claim **1**, wherein the outlet guide vane defines an OGV reference line extending from an inner junction between the outlet guide vane and the turbomachine at a leading edge of the outlet guide vane and an outer junction between the outlet guide vane and the outer nacelle at the leading edge of the outlet guide vane, wherein the turbofan engine defines a radial reference line extending perpendicularly from the longitudinal centerline, and wherein an angle between the OGV reference line and the radial reference line is at least 5 degrees and less than or equal to 45 degrees.
(246) The turbofan engine of any preceding clause, wherein the angle between the OGV reference line and the radial reference line is at least 15 degrees and less than or equal to 35 degrees.
(247) The turbofan engine of any preceding clause, wherein the outer nacelle comprises an outer attachment groove located at a trailing edge of the outlet guide vane.

(248) The turbofan engine of any preceding clause, wherein the turbomachine comprises an inner attachment groove located at the trailing edge of the outlet guide vane, and wherein the inner attachment groove is located aft of the outer attachment groove.
(249) The turbofan engine of any preceding clause, wherein the outer nacelle defines an inlet length, wherein the fan defines a fan diameter, and wherein a ratio of the inlet length to the fan diameter is less than or equal to 0.5.
(250) The turbofan engine of any preceding clause, wherein the outlet guide vane is a first outlet guide vane of a plurality of outlet guide vanes, wherein each outlet guide vane of the plurality of outlet guide vanes defines a base and a tip and is forward swept from the base to the tip.
(251) The turbofan engine of any preceding clause, wherein the turbofan engine defines a bypass ratio equal to a ratio of a mass flowrate of airflow through the bypass passage to a mass flowrate of an airflow through an inlet of the turbomachine during cruise operations, wherein the bypass ratio is at least 5:1 and less than or equal to 20:1.
(252) The turbofan engine of any preceding clause, wherein the fan defines a fan diameter greater than or equal to four feet and less than or equal to 18 feet.
(253) The turbofan engine of any preceding clause, wherein the fan is a single stage fan.
(254) The turbofan engine of any preceding clause, wherein the turbofan defines a maximum rated thrust at sea level greater than or equal to 30,000 pounds and less than or equal to 120,000 pounds.
(255) The turbofan engine of any preceding clause, further comprising: an acoustic treatment attached to or integrated with the outer casing at a location aligned with the outlet guide vane along the longitudinal centerline.
(256) The turbofan engine of any preceding clause, wherein the outlet guide vane defines a junction with the outer casing at a leading edge of the outlet guide vane, and wherein the acoustic treatment is located at the junction, forward of the junction, or both.
(257) A turbofan engine defining an axial direction and a longitudinal centerline along the axial direction, the turbofan engine comprising: a fan section having a fan, the fan comprising a plurality of fan blades; a turbomachine drivingly coupled to the fan, the turbomachine comprising a compressor section with a low pressure compressor, a turbine section with a low pressure turbine, a reduction gearbox, and an outer casing, the low pressure turbine drivingly coupled to the low pressure compressor across the reduction gearbox; an outer nacelle surrounding the fan and at least a portion of the turbomachine; an outlet guide vane extending between the turbomachine and the outer nacelle at a location downstream of the plurality of fan blades, the outlet guide vane defining a base and a tip and being forward swept from the base to the tip.
(258) The turbofan engine of any preceding clause, wherein the low pressure compressor is directly rotatable with the fan.
(259) The turbofan engine of any preceding clause, wherein the reduction gearbox is aligned with the low pressure compressor along the longitudinal centerline.
(260) The turbofan engine of any preceding clause, wherein the outer nacelle defines an inlet length, wherein the fan defines a fan diameter, and wherein a ratio of the inlet length to the fan diameter is less than or equal to 0.5.
(261) The turbofan engine of any preceding clause, wherein the low pressure turbine is a three-stage low pressure turbine.
(262) The turbofan engine of any preceding clause, wherein the compressor section further comprises a high pressure compressor, wherein the high pressure compressor defines a compressor pressure ratio of at least 22 and less than or equal to 30.
(263) The turbofan engine of any preceding clause, wherein the turbomachine comprises an inter-compressor frame, and wherein the base of the outlet guide vane is coupled to the inter-compressor frame.
(264) The turbofan engine of any preceding clause, wherein the base of the outlet guide vane is aligned with the inter-compressor frame along the longitudinal centerline.

(265) The turbofan engine of any preceding clause, wherein the outlet guide vane defines an OGV reference line extending from an inner junction between the outlet guide vane and the turbomachine at a leading edge of the outlet guide vane and an outer junction between the outlet guide vane and the outer nacelle at the leading edge of the outlet guide vane, wherein the turbofan engine defines a radial reference line extending perpendicularly from the longitudinal centerline, and wherein an angle between the OGV reference line and the radial reference line is at least 5 degrees and less than or equal to 45 degrees.
(266) The turbofan engine of any preceding clause, wherein the angle between the OGV reference line and the radial reference line is at least 15 degrees and less than or equal to 35 degrees.
(267) The turbofan engine of any preceding clause, wherein the outer nacelle comprises an outer attachment groove located at a trailing edge of the outlet guide vane.
(268) The turbofan engine of any preceding clause, wherein the turbomachine comprises an inner attachment groove located at the trailing edge of the outlet guide vane, and wherein the inner attachment groove is located aft of the outer attachment groove.
(269) The turbofan engine of any preceding clause, wherein the outer nacelle defines an inlet length, wherein the fan defines a fan diameter, and wherein a ratio of the inlet length to the fan diameter is less than or equal to 0.5.
(270) The turbofan engine of any preceding clause, wherein the outlet guide vane is a first outlet guide vane of a plurality of outlet guide vanes, wherein each outlet guide vane of the plurality of outlet guide vanes defines a base and a tip and is forward swept from the base to the tip.
(271) The turbofan engine of any preceding clause, wherein the turbofan engine defines a bypass ratio equal to a ratio of a mass flowrate of airflow through the bypass passage to a mass flowrate of an airflow through an inlet of the turbomachine during cruise operations, wherein the bypass ratio is at least 5:1 and less than or equal to 20:1.
(272) The turbofan engine of any preceding clause, wherein the fan defines a fan diameter greater than or equal to four feet and less than or equal to 18 feet.
(273) The turbofan engine of any preceding clause, wherein the fan is a single stage fan.
(274) The turbofan engine of any preceding clause, wherein the turbofan defines a maximum rated thrust at sea level greater than or equal to 30,000 pounds and less than or equal to 120,000 pounds.
(275) The turbofan engine of any preceding clause, further comprising: an acoustic treatment attached to or integrated with the outer casing at a location aligned with the outlet guide vane along the longitudinal centerline.
(276) The turbofan engine of any preceding clause, wherein the outlet guide vane defines a junction with the outer casing at a leading edge of the outlet guide vane, and wherein the acoustic treatment is located at the junction, forward of the junction, or both.
(277) This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A turbofan engine defining an axial direction and a longitudinal centerline along the axial direction, the turbofan engine comprising: a fan section having a fan, the fan comprising a plurality of fan blades; a turbomachine drivingly coupled to the fan, the turbomachine comprising a compressor section with a low pressure compressor, a turbine section with a low pressure turbine, a

reduction gearbox, and an outer casing, the low pressure turbine drivingly coupled to the low pressure compressor across the reduction gearbox; an outer nacelle surrounding the fan and at least a portion of the turbomachine; an outlet guide vane extending between the turbomachine and the outer nacelle at a location downstream of the plurality of fan blades, the outlet guide vane having a leading edge and a trailing edge, the outlet guide vane defining a base and a tip and being forward swept from the base to the tip, the tip radially outward of the base; and wherein an aft end of the reduction gearbox is aligned with a downstream most stage of the low pressure compressor along the longitudinal centerline, and wherein the leading edge at the base is located downstream of the aft end of the reduction gearbox along the longitudinal centerline and the trailing edge at the tip is located upstream of the aft end of the reduction gearbox along the longitudinal centerline.

2. The turbofan engine of claim 1, wherein the low pressure compressor is directly rotatable with the fan.

3. The turbofan engine of claim 1, wherein the outer nacelle defines an inlet length, wherein the fan defines a fan diameter, and wherein a ratio of the inlet length to the fan diameter is less than or equal to 0.5.

4. The turbofan engine of claim 1, wherein the low pressure turbine is a three-stage low pressure turbine.

5. The turbofan engine of claim 1, wherein the compressor section further comprises a high pressure compressor, wherein the high pressure compressor defines a compressor pressure ratio of at least 22 and less than or equal to 30.

6. The turbofan engine of claim 1, wherein the turbomachine comprises an inter-compressor frame, and wherein the base of the outlet guide vane is coupled to the inter-compressor frame.

7. The turbofan engine of claim 1, wherein the base of the outlet guide vane is aligned with an inter-compressor frame along the longitudinal centerline.

8. The turbofan engine of claim 1, wherein the outlet guide vane defines an OGV reference line extending from an inner junction between the outlet guide vane and the turbomachine at the leading edge of the outlet guide vane and an outer junction between the outlet guide vane and the outer nacelle at the leading edge of the outlet guide vane, wherein the turbofan engine defines a radial reference line extending perpendicularly from the longitudinal centerline, and wherein an angle between the OGV reference line and the radial reference line is at least 5 degrees and less than or equal to 45 degrees.

9. The turbofan engine of claim 8, wherein the angle between the OGV reference line and the radial reference line is at least 15 degrees and less than or equal to 35 degrees.

10. The turbofan engine of claim 1, wherein the outer nacelle comprises an outer attachment groove located at the trailing edge of the outlet guide vane; and wherein the turbomachine comprises an inner attachment groove located at the trailing edge of the outlet guide vane, and wherein the inner attachment groove is located aft of the outer attachment groove.

11. The turbofan engine of claim 1, wherein the outlet guide vane is a first outlet guide vane of a plurality of outlet guide vanes, wherein each outlet guide vane of the plurality of outlet guide vanes defines a base and a tip and is forward swept from the base to the tip.

12. The turbofan engine of claim 1, wherein the turbofan engine defines a bypass ratio equal to a ratio of a mass flowrate of airflow through a bypass passage to a mass flowrate of an airflow through an inlet of the turbomachine during cruise operations, wherein the bypass ratio is at least 5:1 and less than or equal to 20:1.

13. The turbofan engine of claim 12, wherein the fan defines a fan diameter greater than or equal to four feet and less than or equal to 18 feet, wherein the turbofan engine defines a maximum rated thrust at sea level greater than or equal to 30,000 pounds and less than or equal to 150,000 pounds.

14. The turbofan engine of claim 12, wherein the fan is a single stage fan.

15. The turbofan engine of claim 1, further comprising: an acoustic treatment attached to or integrated with the outer casing at a location aligned with the outlet guide vane along the

longitudinal centerline.
16. The turbofan engine of claim 1, wherein the turbomachine comprises an inter-compressor frame.
17. A turbofan engine defining an axial direction, a longitudinal centerline along the axial direction, and a radial reference line extending perpendicularly from the longitudinal centerline, the turbofan engine comprising: a fan section having a fan, the fan comprising a plurality of fan blades; a turbomachine drivingly coupled to the fan, the turbomachine comprising a compressor section with a low pressure compressor, a turbine section with a low pressure turbine, a reduction gearbox, and an outer casing, the low pressure turbine drivingly coupled to the low pressure compressor across the reduction gearbox, wherein an aft end of the reduction gearbox is aligned with a downstream most stage of the low pressure compressor along the longitudinal centerline; an outer nacelle surrounding the fan and at least a portion of the turbomachine; and a plurality of outlet guide vanes, each outlet guide vane of the plurality of outlet guide vanes extending between the turbomachine and the outer nacelle at a location downstream of the plurality of fan blades, each outlet guide vane of the plurality of outlet guide vanes having a leading edge and a trailing edge, defining a base and a tip and being forward swept from the base to the tip, the tip radially outward of the base, and wherein the leading edge at the base is located downstream of the aft end of the reduction gearbox along the longitudinal centerline and the trailing edge at the tip is located upstream of the aft end of the reduction gearbox along the longitudinal centerline.
18. The turbofan engine of claim 17, wherein each of the plurality of the outlet guide vanes defines an outlet guide vane reference line defined by the leading edge of the respective outlet guide vane, wherein an angle formed between the radial reference line and the outlet guide vane reference line of each respective outlet guide vane is a forward swept angle, wherein the forward swept angle for each outlet guide vane of the plurality of outlet guide vanes is equal to one another.